Overview

* MC-equivalent normalization · ≥ observed calls · † first-order norm propagation may be unreliable

Finite coefficient ε^0 · 25-core integration

Massive bubble

B0(16;3,3), μ^2=1 · average work target 30M / sector

Prescription	Sampler	Generation s	Gen. RAM GB	Artifacts MB	Sectors	f64 max / avg µs / sample	Int. RAM GB	Σ f64 / checked µs / sweep	Average work rel. σ(|I|)*	C	max / |I|	Sector calls 600 s · million	Rel. σ(|I|) 600 s*
Fixed	QMC	0.886	0.0189	0.0178	1	0.149 / 0.149	0.297	0.149 / 0.246	1.88e-17	1.27	3.19	344	**6.26e-18**
	MC						0.658		0.000114	1.27	1.8	247	**4e-05**
Dynamical	QMC	0.861	0.0253	0.0246	1	0.291 / 0.291	0.316	0.291 / 0.389	2.96e-16	2.81	7.91	312	**9.88e-17**
	MC						0.659		0.000272	2.81	4.64	247	**9.47e-05**
Sign-aware	QMC	0.865	0.0508	0.0245	1	0.298 / 0.298	0.567	0.298 / 0.393	2.95e-16	2.81	7.91	338	**9.85e-17**
	MC						0.61		0.00027	2.81	4.64	218	**0.0001**
No contour	QMC	0.542	0.0569	0.0675	6	0.0768 / 0.0716	0.377	0.429 / 1	9.41e-17	1.27	1.07	243	**7.18e-17**
	MC						0.429		9.24e-05	1.27	2.74	114	**0.000118**

Massive triangle

C0(0,0,5;1,1,1), μ^2=1 · average work target 80M / sector

Prescription	Sampler	Generation s	Gen. RAM GB	Artifacts MB	Sectors	f64 max / avg µs / sample	Int. RAM GB	Σ f64 / checked µs / sweep	Average work rel. σ(|I|)*	C	max / |I|	Sector calls 600 s · million	Rel. σ(|I|) 600 s*
Fixed	QMC	1.01	0.0508	0.0307	2	0.17 / 0.167	0.323	0.334 / 0.536	2.7e-10	1.35	17.1	333	**2.08e-10**
	MC						0.642		5.71e-05	1.35	4.21	252	**4.54e-05**
Dynamical	QMC	1.01	0.0191	0.0466	2	0.341 / 0.34	0.316	0.681 / 0.887	8.62e-12	1.26	5.66	312	**6.13e-12**
	MC						0.636		5.21e-05	1.26	2.23	252	**4.18e-05**
Sign-aware	QMC	1.01	0.0476	0.0466	2	0.343 / 0.341	0.323	0.682 / 0.896	8.63e-12	1.26	5.66	339	**6.14e-12**
	MC						0.639		5.29e-05	1.26	2.23	252	**4.2e-05**
No contour	QMC	0.692	0.24	0.35	18	0.0974 / 0.0729	0.443	1.31 / 3.09	5.18e-13	1.58	2.16	340	**9.06e-13**
	MC						0.389		3.1e-05	1.58	1.79	61.9	**0.000151**

Massive box

D0(0,0,0,0,5,−1;1,1,1,1,1), μ^2=1 · average work target 70M / sector

Prescription	Sampler	Generation s	Gen. RAM GB	Artifacts MB	Sectors	f64 max / avg µs / sample	Int. RAM GB	Σ f64 / checked µs / sweep	Average work rel. σ(|I|)*	C	max / |I|	Sector calls 600 s · million	Rel. σ(|I|) 600 s*
Fixed	QMC	1.16	0.0633	0.0442	2	0.209 / 0.205	0.326	0.41 / 0.622	0.000112	2.23	216	369	**5.68e-05**
	MC						0.661		0.000183	2.23	26.2	251	**0.000139**
Dynamical	QMC	1.16	0.0254	0.0705	2	0.407 / 0.399	0.323	0.799 / 1.01	1.83e-08	1.63	29.3	343	**1.02e-08**
	MC						0.673		0.000114	1.63	6.45	255	**8.5e-05**
Sign-aware	QMC	1.16	0.0601	0.0705	2	0.407 / 0.4	0.326	0.801 / 1.02	1.83e-08	1.63	29.3	334	**1.02e-08**
	MC						0.638		0.000115	1.63	6.31	255	**8.48e-05**
No contour	QMC	1.8	0.0949	2.36	54	0.161 / 0.114	0.693	6.13 / 12.5	2.27e-10	4.79	16.5	305	**5.33e-10**
	MC						0.506		8.41e-05	4.79	4.99	29.4	**0.000973**

Overview

* MC-equivalent normalization · ≥ observed calls · † first-order norm propagation may be unreliable

Finite coefficient ε^0 · 25-core integration

Massless sunrise

$p^2=1$; all three masses zero · average work target 60M / sector

Prescription	Sampler	Generation s	Gen. RAM GB	Artifacts MB	Sectors	f64 max / avg µs / sample	Int. RAM GB	Σ f64 / checked µs / sweep	Average work rel. σ(|I|)*	C	max / |I|	Sector calls 600 s · million	Rel. σ(|I|) 600 s*
Fixed	QMC	1.43	0.0886	0.0238	1	0.16 / 0.16	0.31	0.16 / 0.26	4.01e-13	1.01	5.66	153	**2.7e-13**
	MC						0.358		2.58e-05	1.01	3.21	120	**1.84e-05**
Dynamical	QMC	1.49	0.0887	0.027	1	0.157 / 0.157	0.319	0.157 / 0.257	4.04e-13	1.01	5.66	149	**2.7e-13**
	MC						0.362		2.6e-05	1.01	3.46	126	**1.81e-05**
Sign-aware	QMC	1.45	0.0728	0.0242	1	0.162 / 0.162	0.313	0.162 / 0.263	4.07e-13	1.01	5.66	151	**2.71e-13**
	MC						0.37		2.58e-05	1.01	3.07	130	**1.77e-05**
No contour	QMC	0.539	0.0254	0.104	6	0.187 / 0.179	0.501	1.08 / 1.66	1.67e-13	1.01	1.19	306	**2.23e-13**
	MC						0.4		1.05e-05	1.01	0.426	106	**1.94e-05**

Massive sunrise

m1=m2=m3=1, s=16; native measure1 · average work target 0.323M / sector

Prescription	Sampler	Generation s	Gen. RAM GB	Artifacts MB	Sectors	f64 max / avg µs / sample	Int. RAM GB	Σ f64 / checked µs / sweep	Average work rel. σ(|I|)*	C	max / |I|	Sector calls 600 s · million	Rel. σ(|I|) 600 s*
Fixed	QMC	1.61	0.0444	0.0365	1	0.652 / 0.652	0.485	0.652 / 1.74	1.07e-11	1.25	5.88	232	**5.27e-13**
	MC						0.635		0.00097	1.25	9.83	213	**3.78e-05**
Dynamical	QMC	1.44	0.0381	0.0801	1	3.62 / 3.62	0.568	3.62 / 7.09	8.75e-10	3.84	23.7	160	**4.29e-11**
	MC						0.608		0.00545	3.84	49.5	207	**0.000216**
Sign-aware	QMC	1.44	0.0413	0.0859	1	2.47 / 2.47	0.568	2.47 / 5.12	4.38e-09	3.87	24.1	154	**2.15e-10**
	MC						0.628		0.00548	3.87	45.8	202	**0.00022**
No contour	QMC	6.7	0.215	2.3	62	2.79 / 0.764	0.497	47.4 / 2.37e+04	3.65e-11	1.54	4.15	25.9	**3.95e-11**
	MC						0.561		0.000301	1.54	1.66	14.8	**0.000351**

Overview

* MC-equivalent normalization · ≥ observed calls · † first-order norm propagation may be unreliable

Finite coefficient ε^0 · 25-core integration

Massive-threshold kite

p^2=16; five internal masses equal to 1 · average work target 40M / sector

Prescription	Sampler	Generation s	Gen. RAM GB	Artifacts MB	Sectors	f64 max / avg µs / sample	Int. RAM GB	Σ f64 / checked µs / sweep	Average work rel. σ(|I|)*	C	max / |I|	Sector calls 600 s · million	Rel. σ(|I|) 600 s*
Fixed	QMC	2.98	0.238	0.59	4	0.445 / 0.42	0.326	1.68 / 2.14	2.71e-05	1.86	17.2	377	**1.82e-05**
	MC						0.599		0.000125	1.86	11.8	205	**0.000111**
Dynamical	QMC	3.93	0.275	4.05	4	4.13 / 3.1	0.405	12.4 / 12.9	0.000164	2.5	57.8	353	**0.000147**
	MC						0.672		0.00065	2.5	3.03e+03	209	**0.000408**
Sign-aware	QMC	5.18	0.319	7.01	4	4.76 / 3.68	0.483	14.7 / 15.2	0.000185	2.6	59.8	351	**0.000172**
	MC						0.641		0.000518	2.59	1.23e+04	207	**0.000853**
No contour	QMC	Not completed	—	—	—	—	—	—	—	—	—	—	—
	MC						—		—	—	—	—	—

Overview

* MC-equivalent normalization · ≥ observed calls · † first-order norm propagation may be unreliable

Scalar D05 double box

s=1000000, t=−97077, mH=125, mt=345/2 · average work target 4M / sector

Prescription	Sampler	Generation s	Gen. RAM GB	Artifacts MB	Sectors	f64 max / avg µs / sample	Int. RAM GB	Σ f64 / checked µs / sweep	Average work rel. σ(|I|)*	C	max / |I|	Sector calls 600 s · million	Rel. σ(|I|) 600 s*
Fixed	**QMC**	9.77	0.561	28.8	18	0.738 / 0.724	0.475	13 / 15.1	0.0259	9.95	7.02e+03	363	**0.0102**
	MC						0.653		0.00256	9.98	740	85.3	**0.00246**
Dynamical	**QMC**	19.2	1.46	77.8	18	5.77 / 4.68	0.64	84.3 / 87	0.306†	19.8	1.43e+07	342	**0.169†**
	MC						0.837		0.286†	14.3	1.09e+06	67.1	**0.167†**
Dynamic tuned	**QMC**	19.2	1.46	77.8	18	6.2 / 5.33	0.956	95.9 / 98.5	0.0118	9.37	1.75e+03	289	**0.00665**
	MC					5.14 / 4.4	0.862	79.2 / 81.8	0.00271	10	692	77.3	**0.00262**
Sign-aware	**QMC**	34.1	2.03	88.6	18	6.74 / 5.47	1.13	98.5 / 101	0.328†	24.4	4.04e+06	326	**0.197†**
	MC						0.965		0.626†	35.2	5.04e+06	82.5	**0.586†**
Sign-aware tuned	**QMC**	34.1	2.03	88.6	18	7.36 / 6.16	1.08	111 / 114	0.0125	10.7	1.98e+03	284	**0.00755**
	MC					6.16 / 5.23	0.997	94.2 / 96.7	0.00248	10	546	82.5	**0.00233**

Overview

* MC-equivalent normalization · ≥ observed calls · † first-order norm propagation may be unreliable

Physical D05 double box

s=1000000, t=−97077, mH=125, mt=345/2; incoming ++ · average work target 6M / sector

Prescription	Sampler	Generation s	Gen. RAM GB	Artifacts MB	Sectors	f64 max / avg µs / sample	Int. RAM GB	Σ f64 / checked µs / sweep	Average work rel. σ(|I|)*	C	max / |I|	Sector calls 600 s · million	Rel. σ(|I|) 600 s*
Fixed	**QMC**	253	8.99	38.1	30	3.22 / 3.01	0.476	90.4 / 94.5	0.0311	23.1	1.04e+04	344	**0.0198**
	MC						0.755		0.00329	23.2	1.03e+03	44.6	**0.00594**
Dynamical	**QMC**	1.8e+03	34.8	88.6	30	8.08 / 6.93	0.54	208 / 213	0.336†	44.5	2.81e+08	328	**0.457†**
	MC						1.09		0.308†	200	6.95e+05	44.2	**1.95†**
Dynamic tuned	**QMC**	1.8e+03	34.8	88.6	30	7.52 / 6.71	0.775	201 / 206	0.0272	22.6	9.79e+03	297	**0.0209**
	MC					7.52 / 6.71	1.14	201 / 206	0.00324	23.1	1.43e+03	43.6	**0.00662**
Sign-aware	**QMC**	1.54e+03	36.7	99.7	30	9.11 / 7.72	0.805	232 / 237	0.924†	98.8	8.71e+07	332	**0.708†**
	MC						1.2		0.333†	61.6	7.27e+05	43.6	**0.68†**
Sign-aware tuned	**QMC**	1.54e+03	36.7	99.7	30	8.98 / 7.59	0.807	228 / 232	0.0287	22.8	9.69e+03	323	**0.0214**
	MC					8.98 / 7.59	1.2	228 / 232	0.00313	23	1.12e+03	42.4	**0.00648**

1 Overview: scope and reproduction

Final campaign scope. The overview contains 66 measured timed sampler rows across eight topologies, retaining the original 44 arms and separately identified extensions. The user removed double-box no-contour development from scope. Kite no-contour remains not completed; its boundary milestone and substantial remaining work are documented below. A dash is an unavailable measurement, never a zero. The opening tables use the new finite-coefficient campaign only. Historical studies below retain their original points, settings and resources; they are not substituted for missing new measurements.

All uncertainties target ε^0 of the complete complex integral. Lower Laurent coefficients and their full covariance are retained. The primary final column is the relative standard error of the **magnitude** $M = |I_0|$. Writing $I_0 = a + ib$, first-order propagation gives

$$\sigma_M^2 = \frac{a^2 C_{RR} + 2ab C_{RI} + b^2 C_{II}}{M^2}.$$

The relative error is $\frac{\sigma_M}{M}$. Zero magnitude makes this undefined; large uncertainty relative to the magnitude receives a linearization warning. The dagger marks joint complex SE divided by magnitude at least 0.1; this is a warning convention, not a certified validity boundary. This radial error measures the modulus, not accuracy in its complex phase. The joint complex SE $\sqrt{C_{RR} + C_{II}}$ remains in the detailed data and scientific compatibility checks. Here C is the covariance of the estimated mean, not of individual sample weights. Neither SE is an independently established error against an exact result. The discrete-MC total uses its native full-batch covariance; summing sector marginal variances would lose its cross-sector correlations.

The bubble QMC estimates reach the scale of floating-point rounding. Their very small reported replica SE does not include common numerical bias or imply accuracy beyond f64. Its independent finite reference is $2 - 3\frac{\log(3)}{2} + i\frac{\pi}{2}$; reference discrepancies are kept separate from sampling SE.

1.1 Clocks, work and memory

Generation is launch-to-durable-publication wall time, excluding the software build. Artifact size counts the manifest and its indexed data exactly once. Sector count means executable stochastic kernels; preparation charts and exact contributions have separate native records. Generation and integration RAM are sampled aggregate resident memory of each owned process tree, in decimal GB. Shared mappings can be counted more than once and sampling can miss peaks; these numbers are not a measurement of unique physical pages.

f64 max / avg means the maximum and arithmetic mean of per-sector mean kernel costs, not the slowest individual point. **Σ f64 / checked** sums those means for one complete sweep and separately sums the per-sector measured wall costs with normal stability checks and precision rescue. It is not a 25-core wall-time prediction. The native profiling harness uses deterministic coordinates (4096 for the original smaller scalar cases, 1024 for both D05 variants and the new massive-sunrise/bubble extensions), three repetitions and zero warmup; call counts and coordinate counts are retained separately. Loading and evaluator restoration have their own receipts. The integration elapsed clock starts after ordinary coordinator kernel loading; serial resident loading and subsequent reloading occur inside its clock. Process-wall and loading times are retained separately so these startup costs are not confused with the reported sampling interval.

Causal validation is off throughout every new table timing: profiling, throughput calibration, matched-work integration and the 600-second arm all request this explicitly. No causal pilot is included. The collector rejects contour receipts with a different policy or a nonzero causal-check count. Mandatory solver termination, nonfinite handling and precision rescue remain active. Earlier checked-policy measurements are separate diagnostic evidence.

Independent throughput pilots choose the work allocations and the round average-per-sector targets listed in the data. Discrete MC samples sectors unequally; average work is the accepted total divided by the number of stochastic sectors, not a claim that each sector has the same allocation. The secondary display rescales the radial relative SE by $\sqrt{\frac{N_{\text{actual}}}{N_{\text{target}}}}$. The primary time display rescales it by $\sqrt{\frac{t_{\text{actual}}}{600}}$. Both corrections are **Monte-Carlo-equivalent heuristics**, not measured QMC convergence laws. Actual errors, counts, factors and times are retained unchanged. A topology's time comparison is withheld if its longest/shortest accepted run exceeds 1.25. Each accepted timed run must also lie between 480 and 750 seconds. Each confirmation has a seed disjoint from calibration and the other arm. The instrumented fixed-time arm stops new reservations at 600 native seconds and drains complete in-flight tasks or batches before saving. Its exact call count requires a clean native drain. Earlier confirmations used graceful cancellation and retain their observed-prefix counts. Actual elapsed time, drain and process-wall time remain separate. An externally interrupted run does not qualify for the new timed table. The integration-RAM column uses this arm.

The secondary work display may use a clean capped estimate with an unfinished planned allocation, provided its native full-coverage covariance comes from committed complete batches under the same fixed design. Its strict matched-work status remains incomplete; the actual count and square-root correction are retained explicitly. A completed continuation is used instead of its partial prefix, without pooling the two estimates. For the new triangle no-contour secondary arm, QMC completed all 241,606,656 planned evaluations in 402.311 seconds; MC saved 40,550,400 at its clean 900-second deadline (904.709 seconds including drain). That MC allocation remains incomplete. Neither secondary time is described as a 600-second measurement. The new sunrise no-contour secondary arms both completed their pilot-selected 26,566,656 evaluations, in 59.851 seconds for QMC and 122.960 seconds for MC. Their correction to the same round work target is explicitly heuristic; the separate timed confirmations still run for approximately 600 seconds.

The new box no-contour QMC secondary arm completed 161,243,136 planned evaluations in 319.459 seconds. MC saved 19,148,800 at its clean 900-second deadline (908.258 seconds including drain); that planned allocation remains incomplete. Its actual-work display correction is only the same square-root heuristic. The two independent timed arms remain near 600 seconds.

The timed-call column counts observed logical sector-integrand evaluations, including reported pilot and discarded work. Precision-rescue backend calls are retained separately and do not become new sampling points. A leading $\geq$ identifies an older lower bound: process cancellation may lose the unpublished tail since a worker's last status message. Unprefixed counts require complete native deadline/drain accounting. Accepted production work is recorded separately and exactly. New weight-diagnostic runs require their own complete accounting; no total is inferred from elapsed time or throughput.

The cancellation and maximum-weight columns use fresh native accepted-production accumulators. A dash means that this arm has no accepted instrumented measurement yet; old receipts cannot reconstruct these statistics. Accumulator and observation costs remain inside the new native elapsed clock. The cancellation numerator is the sum of absolute sector integrals plus the magnitude of the combined exact offset, divided by $|I_0|$. The maximum uses actual complex magnitudes, with continuous importance correction retained and discrete sector-selection correction removed for a conditional sector weight. It depends on the proposal and sampled work; it is neither a supremum nor a tail bound.

1.2 Execution and topology definitions

Generation uses serial sector residency with 25 workers for the scalar suite. The physical-numerator double box uses six generation workers to control its larger per-sector memory; this is the explicit generation exception. Integration uses 25 workers for both samplers. QMC uses serial residency with a one-second minimum stay. MC denotes the existing **ordinary discrete-MC scheduler**, the user-authorized exception to serial integration. The shared owned campaign cohort observes a 100 GB allowance. Campaigns may overlap; no quiet-machine assumption is made. The caller now permits up to five hours for a double-box generation attempt; this ceiling is a preparation allowance, not a change to the 600-second integration comparison or a relabelling of any completed measurement.

The input cards in `examples/no_deformation/benchmarks/overview/` specify the native HEPKit graphs, exact masses/invariants, normalization, symbolic endpoint subtraction and order zero. The massive kite has all five masses equal to one and $p^2 = 16$; the historical mixed-mass small-momentum kite is a different case. The massless sunrise has no interior threshold zero and is retained as a non-crossing control. The scalar and physical-numerator D05 double boxes share $s = 10^6$, $t = -97077$, $u = -871673$, $m_H = 125$, $m_t = \frac{345}{2}$. The scalar numerator is one; the physical amplitude retains its original numerator and prefactors.

The smaller scalar cases use fixed $\lambda = 0.1$. The D05 comparison uses $\lambda = 10^{-6}$; its trial value 0.1 failed a separate causal admission check and contributes no accepted benchmark result. Dynamic runs use $S = 0.8$, $L = 1$ and $R = 1$, with polynomial and sign-aware constructions labelled separately. The D05 fixed and polynomial settings passed separate finite pilots; those checks supply sampled safeguards, not a global certificate, and their clocks are excluded from every overview timing.

Symbolic and contour-only dual Jacobians are compared in preparation while endpoint reduction remains symbolic. The selected candidate minimizes the measured full-sweep cost before any confirmation result is available. Full commands, artifact identities, evaluator sizes, component layout and binary digests are retained in `docs/benchmark-overview-results.json`; the protocol and input inventory describe reproduction. These measured binaries use the pinned symGCAD revision `644a560e69a064f811fa93aff33a247ea7fcdfca`. The original 38 instrumented timed arms use executable `cf9eb450`; the subsequently admitted triangle no-contour arms use `35c92289`, with their distinct source/build receipt. The subsequently admitted sunrise no-contour arms use `3cf81e4b`. The subsequently admitted box no-contour arms use `d94dd90d`. The triangle executable retains normal stability checks and precision rescue in checked costs and integration, even when the raw f64 evaluation is much cheaper. Its timed QMC run rescued 115,307,844 of 340,410,368 logical calls (33.9%); the MC run rescued all 61,915,136 calls, with zero failures and maximum precision 256 bits. Their weighted-check / additional-replay counts were 21 / 12 and 144 / 0. The deterministic profiling coordinates triggered no rescue. These are different sampling workloads; the archive has no per-trigger rescue reason

histogram. The checked integration clock retains this overhead. The new sunrise timed QMC run rescued 103,781,710 of 306,384,896 calls (33.9%); MC rescued all 106,426,368 calls. Both reached 256 bits with zero failures. Their weighted-check / additional-replay counts were 8 / 6 and 48 / 0.

The new box timed QMC run rescued 98,390,993 of 305,291,264 calls (32.2%); MC rescued all 29,368,320 calls. Both returned zero failures; maximum precision was 384 and 256 bits, respectively. The checked clocks retain this overhead, independently of the raw f64 profile.

1.3 Current admission limits

The bubble, triangle, massless sunrise, massive sunrise and massive box have complete no-contour integration rows. The triangle uses its unchanged native HEPKit graph input, four freshly verified source cells, and 18 endpoint charts; its quoted generation time includes the complete publication step. Fresh-process loading and separate QMC/MC confirmations use that actual artifact. The earlier unsupported attempt is retained as historical evidence rather than overwritten. The sunrise uses its unchanged native graph, one verified source cell and six endpoint charts. Its separately seeded confirmations agree with the retained analytic identity under the reported full-covariance diagnostics.

The box now uses six verified source cells and 54 endpoint charts, with the original physical graph, point and OneLOop normalization. Its explicit native order changes the CAD presentation only. Full publication, fresh-process loading and separate 600-second QMC/MC confirmations are measured.

The original frozen benchmark CLI reported “Unsupported” for box and kite before invoking GCAD. Those original refusal receipts remain historical. The kite still lacks a complete endpoint artifact and no-contour numerical row. A later native solve and fresh verification produced 9,840 cells. Selected moving-boundary admission now passes, but full interval densities, closed-face continuation and whole-atlas coverage remain unfinished. These are implementation gaps, not rejected physical integrals or evidence of GCAD infeasibility. Geometry-only work is not counted as complete generation. The existing native HEPKit CLI graph-parametrization route binds the prepared source and input fingerprint; it is distinct from the stricter pre-parametrization GraphPoint owner API. Surviving inexact geometric coefficients are refused, not rounded. The user stopped double-box no-contour development after the documented local admission result. Those two methods are omitted from the opening overview; the subsequent boundary discussion records their atlas, chart counts and implementation limits. GCAD-only timings are not complete generation measurements. The studies below retain their original scope and measurements.

Campaign completion update. The massive sunrise now has measurements for all four intended prescriptions and both samplers. The scalar and physical double boxes now both include measured sign-aware results. The bubble now also contains its separate sign-aware confirmation. Previously measured entries retain their original values and run identities.

For massive-kite cell 0, all ten replacement boundary charts pass 40 strict native 128/256-bit checks. This is a selected fourth boundary over three integration directions, not a complete four-dimensional interval density. The remaining 9,839 cells and Laurent continuation make completion substantial work; it is reported as **Not completed** under the final user scope. The double-box scalar diagnostic admitted 307 charts of one original cell, with zero compiled kernels or integral estimates. Neither preparation result is entered as full-integral generation or integration time.

Scalar double-box sign-aware extension. Fresh serial generation with 25 workers took 34.15 seconds and 2.03 GB sampled peak RSS, producing 18 kernels and 88.60 MB of artifacts. Symbolic endpoint reduction and symbolic contour Jacobians use Horner10 and SymJIT O2. The recipe was absent from the previous campaign, rather than rejected by the implementation. The sign-aware bound penalizes higher-order terms that oppose the causal direction, allowing helpful terms without imposing the same absolute-magnitude penalty. It does not imply better variance for every topology.

At the predeclared $S = 0.8$, $L = R = 1$, independent QMC and ordinary discrete-MC confirmations ran for 600.159 and 601.285 native seconds. Their 600-second normalized radial relative errors are 0.197 and 0.586, both carrying the native large-uncertainty warning. These runs show no improvement over the existing polynomial-dynamic measurements; they do not establish an optimal tuning or a high-precision integral value. Sixteen marginal comparisons with the currently reported fixed and polynomial arms pass the stated compatibility diagnostic, with a maximum difference of 1.349 combined standard errors. Full covariance and distinct artifact identities are retained. The secondary target remains four million accepted evaluations per sector, with the explicit square-root display correction. Compact results and exact invocation settings are in `docs/scalar-double-box-sign-aware-overview.json`.

Physical double-box sign-aware extension. Independent full-source generation took 1536.10 seconds and 36.67 GB peak aggregate RSS with six resident generation workers. It produced 30 kernels and 99.71 MB of artifacts, retaining the physical numerator and full four-component Laurent vector. The raw f64 maximum/average sector costs are 9.108/7.724 microseconds; their full-sweep sum is 231.72 microseconds, or 236.72 microseconds with stability handling in this profile.

At the same predeclared $S = 0.8$, $L = R = 1$, QMC made 331.70 million logical calls in 600.135 seconds and ordinary discrete MC made 43.59 million in 603.712 seconds. Their normalized radial relative errors are 0.708 and 0.680, with large-uncertainty warnings. All 32 marginal comparisons with the fixed and polynomial confirmations pass the stated compatibility diagnostic; the largest is 2.643 combined SE. This is a noisy baseline, not an accuracy certificate. The physical compact record is `docs/physical-double-box-sign-aware-overview.json`.

1.4 Double-box dynamic-strength tuning

The new **Dynamic tuned** and **Sign-aware tuned** rows retain the same artifacts as their original counterparts. Their generation costs therefore repeat the original measured receipts; no symbolic regeneration was necessary. The original $L = R = 1$ results remain visible. Sampling-cost profiles were measured again at the selected settings, separately for each sampler when those settings differ.

Independent short calibrations selected $S = 0.8$, displacement cap $R = 10^6$, and strength cap $L = 1.25 \times 10^{-5}$ for scalar QMC. Scalar discrete MC and both physical samplers use $L = 1.25 \times 10^{-6}$. The eight confirmation streams were fixed before their runs and are disjoint from calibration. They use 25 workers, the existing serial QMC/ordinary discrete-MC scheduling, and shared 100 GB memory limits. All runs keep the complete position-dependent strength derivatives, Jacobian and mandatory numerical precision handling. Causal validation is off.

Finite coefficient	Fixed	Dynamic tuned	Sign-aware tuned	Sampler
Scalar double box	1.02%	0.665%	0.755%	QMC
Scalar double box	0.246%	0.262%	0.233%	MC
Physical double box	1.98%	2.09%	2.14%	QMC
Physical double box	0.594%	0.662%	0.648%	MC

These are relative standard errors of the complex magnitude, corrected to 600 seconds with the stated square-root display convention. Native timed durations are 600.11–604.79 seconds. The tuned physical results recover performance close to the fixed map; the scalar QMC results improve on it. This is a demonstrated improvement over the original dynamic settings, not evidence of a universal dynamic advantage. All eight confirmations have complete production coverage, no failed, unstable or discarded-zero points, and no norm-linearization warning.

Why the caps matter. The original displacement bound can control the radius when dimensionful gradients are large. Increasing R while decreasing L changes both the allowed strength and its spatial derivatives. For the polynomial construction, sufficiently small L approaches $\lambda = SL$ at fixed regular coordinates when the other bounds become inactive. The sign-aware smoothing introduces a small limiting correction, retained in the native tests. Neither statement certifies an arbitrary fixed-strength contour globally.

At three native scalar-sector test points, $L = 1.25 \times 10^{-6}$ and $R = 10^6$ give $\frac{\lambda}{SL} = 0.9792, 0.9915, 0.9985$ for the polynomial bound. The Jacobian still includes the strength gradient. At the tested midpoint, the larger scalar QMC cap instead permits displacement 1.20, versus 0.742 in the old setting; it is not simply a copy of the fixed map. These are coordinate diagnostics, not a claim about every sector or the largest observed integration weights. The saved native weight summaries do not retain the coordinates of those maxima.

Independent work controls and compatibility. Three scalar QMC controls each completed exactly 71,958,528 accepted calls. Tuned polynomial/fixed covariance trace is 0.193; sign-aware/fixed is about 0.224. Thus the scalar improvement also appears at equal accepted work, using the full complex error rather than just radial relative uncertainty. These independent estimates are not pooled with the timed runs. Three physical QMC controls completed exactly 179,896,320 accepted calls each: fixed, polynomial and sign-aware took 377.81, 406.92 and 328.12 seconds. Their covariance-trace ratios to fixed are 1, 0.9904 and 0.9790. The physical tuning therefore recovers approximately the fixed per-call variance; these independent single controls do not establish a statistically significant variance advantage. Scalar fixed, polynomial and sign-aware controls took 112.49, 145.95 and 136.80 seconds at their common work target.

All six MC work controls stopped at their original 600-second caps, with 40–41 million scalar or 20–21 million physical accepted calls. They remain explicitly incomplete requested work; there was no retry to force equal counts. All twenty predeclared final jobs are closed without failed, unstable or cutoff-zero evaluations.

For the secondary overview column, completed work controls receive the small explicit correction to 72 million scalar or 180 million physical calls. The fresh fixed QMC controls supply the corresponding secondary baseline values; original timed baseline results remain unchanged. Otherwise the timed estimate is scaled to the designated 72 million scalar or 180 million physical calls using the stated MC-equivalent heuristic. This does not relabel unfinished allocations as equal-work measurements. The final ten-minute column is the primary metric.

Native Numerica matrix checks compare the 12 retained baseline estimates, eight new timed estimates and six completed QMC controls: 234 complex-block comparisons. The largest descriptive Mahalanobis distances are 3.169 for scalar ε^0, 3.805 for physical ε^{-1} and 3.280 for physical ε^0. There are no non-positive covariance or rounding-sensitivity flags in this set. Component checks are also retained. These estimated-covariance diagnostics are not calibrated p-values or an independent analytic reference for this diagram.

The compact record `docs/double-box-dynamic-tuning-overview.json` preserves exact native arguments, independent seeds, elapsed times, accepted/total calls, full Laurent means and covariance, C, maximum-weight ratios, memory and source identities. The completed work controls and final 234-pair audit are in `docs/double-box-dynamic-work-overview.json`. Both use frozen executable source `6647950` and the original D05 graph, model, parameters and generation cards. Only runtime contour caps change.

Bubble sign-aware completion. A fresh 0.865-second generation produces one kernel. Independent QMC/MC confirmations take 600.043/602.843 seconds and give normalized radial relative errors $9.85 \times 10^{-17}/1.00 \times 10^{-4}$. The first value is at floating-point roundoff and is not a numerical accuracy claim. Eight full-vector marginal checks against the retained independent reference pass, with a maximum 1.203 combined SE. See `docs/bubble-sign-aware-overview.json` for exact settings and native results.

Massive-sunrise extension (12 October). The new opening block is a distinct complete source with all three masses one and $s = 16$, native measure one and $D = 4 - 2\varepsilon$. Release `6647950` generated all 62 charts in 6.70 seconds, including the fresh solve and automatic endpoint admission. Horner10, CPE1000 and SymJIT O2 were used. QMC has one-second minimum serial residence; MC uses the authorized ordinary scheduler. Both use 25 physical cores, Korobov3, 4096-point initial designs, eight initial replicas, fixed-size continuation, validated algebraic precision rescue and optional causal checks off.

Independent confirmation seeds 202610125021 and 202610125022 ran for 604.97 and 602.67 seconds including native task drain. The preceding 30-second calibrations are not pooled. The secondary work display scales to 20 million accepted calls in total (20 million / 62 per sector); it is a square-root heuristic from these same timed runs, not an additional matched-work experiment. The final column uses native radial norm uncertainty and the small square-root correction to 600 seconds. Complete covariance, accepted counts, total calls, diagnostic timings and settings are in `docs/no-contour-massive-sunrise-overview.json`. Fresh matching ten-minute fixed, polynomial-dynamic and sign-aware contour arms now appear alongside it; earlier short controls are not substituted. The portable reproduction card is `examples/no_deformation/benchmarks/controls/massive_sunrise_no_contour.toml`; commands and measurement scope are in `docs/benchmark-overview-protocol.md`.

The six contour confirmations use the identical frozen executable, full Laurent orders $-2, -1, 0$, symbolic endpoint reduction and symbolic Jacobians. Their generation times are 1.61, 1.44 and 1.44 seconds, with one compiled kernel per recipe. Fixed $\lambda = 0.1$ gives relative norm SE of 5.27×10^{-13} (QMC) and 3.78×10^{-5} (MC) after the small 600-second correction. Polynomial dynamics gives 4.29×10^{-11} and 2.16×10^{-4}; sign-aware dynamics gives 2.15×10^{-10} and 2.20×10^{-4}. All dynamic settings remain the predeclared $S = 0.8$, $L = R = 1$.

The secondary average-work target is the same $20 \times \frac{10^6}{62}$ evaluations per sector for every method, so a one-kernel contour arm is rescaled to that total rather than to 20 million. This large work rescaling remains a heuristic; the near-ten-minute column is the primary comparison. Native covariance, diagnostics, settings, exact generation metrics and all six fresh results are in `docs/massive-sunrise-contour-overview.json`.

Thirty-six marginal comparisons with the independently generated pySecDec Laurent reference pass, with maximum discrepancy 2.029 combined SE. The existing native Numerica matrix diagnostic additionally compares all eight campaign arms over all three complex Laurent blocks: 84 comparisons,

largest defined descriptive Mahalanobis distance 2.611. Five pole comparisons are rounding-sensitive and two covariance blocks are not positive definite; their diagnostics are retained without regularization or inflated errors. Estimated QMC replica SE does not bound common numerical bias.

Topology	Prescription	Measurement status
Massive bubble	Fixed	New epsilon-zero serial generation (25 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Massive bubble	Dynamical	New epsilon-zero serial generation (25 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Massive bubble	Sign-aware	Fresh same6647950 symbolic-Jacobian generation/cost and independent600s confirmations; pilots unpooled; full native Laurent covariance retained.
Massive bubble	No contour	New epsilon-zero serial generation (25 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Massive triangle	Fixed	New epsilon-zero serial generation (25 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Massive triangle	Dynamical	New epsilon-zero serial generation (25 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Massive triangle	Sign-aware	New epsilon-zero serial generation (25 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Massive triangle	No contour	New complete native graph threshold route: four verified source cells and 18 endpoint charts; prior unsupported attempt remains historical.
Massive box	Fixed	New epsilon-zero serial generation (25 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Massive box	Dynamical	New epsilon-zero serial generation (25 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Massive box	Sign-aware	New epsilon-zero serial generation (25 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Massive box	No contour	Six verified source cells and 54 endpoint charts; explicit native order [v1, v2, v0], full publication and separate confirmations. Prior unsupported attempt retained.
Massless sunrise	Fixed	New epsilon-zero serial generation (25 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Massless sunrise	Dynamical	New epsilon-zero serial generation (25 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Massless sunrise	Sign-aware	New epsilon-zero serial generation (25 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Massless sunrise	No contour	New complete native graph threshold route: one verified source cell and six endpoint charts; prior unsupported attempt remains historical.
Massive-threshold kite	Fixed	New epsilon-zero serial generation (25 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Massive-threshold kite	Dynamical	New epsilon-zero serial generation (25 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Massive-threshold kite	Sign-aware	New epsilon-zero serial generation (25 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Massive-threshold kite	No contour	Verified 9,840-cell GCAD atlas. Ten selected-boundary charts of cell 0 pass 40 strict precision checks; complete interval/source-density closure and whole-atlas continuation remain substantial work. No full-integral artifact or numerical result. Stopped under the final user scope.
Scalar D05 double box	Fixed	New epsilon-zero serial generation (25 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Scalar D05 double box	Dynamical	New epsilon-zero serial generation (25 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Scalar D05 double box	Dynamic tuned	Frozen sampler-specific caps selected with independent calibration; fresh 600 s confirmations and separate bounded work controls; full native Jacobian/precision retained.
Scalar D05 double box	Sign-aware	Fresh complete scalar double-box symbolic generation and independent600s QMC/ordinaryMC; optional causal checks off; native norm warnings retained.
Scalar D05 double box	Sign-aware tuned	Frozen sampler-specific caps selected with independent calibration; fresh 600 s confirmations and separate bounded work controls; full native Jacobian/precision retained.
Physical D05 double box	Fixed	New epsilon-zero serial generation (6 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Physical D05 double box	Dynamical	New epsilon-zero serial generation (6 resident workers) and native Off-policy cost measurement; integration status shown per sampler
Physical D05 double box	Dynamic tuned	Frozen sampler-specific caps selected with independent calibration; fresh 600 s confirmations and separate bounded work controls; full native Jacobian/precision retained.

Topology	Prescription	Measurement status
Physical D05 double box	Sign-aware	Fresh complete physical double-box symbolic generation and independent600s QMC/ordinaryMC; optional causal checks off; native norm warnings retained.
Physical D05 double box	Sign-aware tuned	Frozen sampler-specific caps selected with independent calibration; fresh 600 s confirmations and separate bounded work controls; full native Jacobian/precision retained.
Massive sunrise	Fixed	Fresh same6647950 symbolic-Jacobian generation/cost and independent600s confirmations; pilots unpooled; full native Laurent covariance retained.
Massive sunrise	Dynamical	Fresh same6647950 symbolic-Jacobian generation/cost and independent600s confirmations; pilots unpooled; full native Laurent covariance retained.
Massive sunrise	Sign-aware	Fresh same6647950 symbolic-Jacobian generation/cost and independent600s confirmations; pilots unpooled; full native Laurent covariance retained.
Massive sunrise	No contour	Complete62-chart no-contour generation and independent600s confirmations; now compared with fresh fixed/polynomial/sign-aware arms on same frozen executable.

2 No-contour boundary limits and final scope

The double-box no-contour work ended before either the scalar or physical integral was generated; both were removed from the requested support and benchmark scope. The available contour comparisons remain separate. The massive-kite work reached a selected-boundary milestone, but no complete interval density or full-integral benchmark.

2.1 Algebraic boundaries and admitted local charts

The verified double-box decomposition contains **644,699 original generic-open cells**. Its internal monotone base has 464,265 cells; that is not a smaller complete atlas. After the positive projective anchor x_0, the retained lifting order and exact catalogue complexity are:

Variable	Prefix nodes	Boundary polynomials	Maximum degree
x_1	416	222	32
x_3	10,543	24	4
x_4	102,146	7	4
x_5	334,779	3	2
x_6	464,265	2	1
x_2	644,699	2	2

Degrees refer to the boundary polynomial in its current lifting variable, not a specialized minimal polynomial. Later roots move with earlier coordinates. Their gaps, vanishing leading coefficients, collisions, infinity boundaries and intersections of closed faces require more than a constant-root solver or a sampled sign check.

Native general improvements addressed parts of this problem: source-owned constant endpoints through degree 32; exact coefficient extraction for tight dyadic root brackets; complete recursive fan replacements; original-certificate replay under checked positive rational scaling; and canonicalization of the full composed measure while retaining source denominators and ancestry. They preserve selected branches, causal phases and the Jacobian exactly once. These operations reuse Symbolica algebra and symGCAD certificates; they do not constitute an automatic resolver for every moving boundary.[1]

For original double-box cell 0, native source-bound admission completed **307 closed scalar charts**, retaining the full regulator density and the common sufficient strip $-\frac{3}{2} < \Re(\varepsilon) < 1$. This produced **zero compiled kernels and zero integral estimates**. The physical numerator was not admitted by this scalar test. R5 took **4,122.9 s** and peaked at **9.96 GB own aggregate RSS**, with one proof decode and fresh verification followed by caller-owned replay work. It was a **debug preparation diagnostic**, not release CLI generation of a full integral. The admitted objects ended with the process; saved receipts alone do not recreate their source authority.

2.2 Why catalogue sharing is not complete map coalescence

Exact prefix bookkeeping reduces 3,868,194 per-cell axis occurrences to 1,556,848 nodes, a factor of 2.485. Repeated root-bound uses similarly reduce by a factor of 3.003. The 260 distinct global boundary polynomials and 45 global support classes do not establish equality of selected branches, transformed factors, maps or closed domains. Native compatibility merging retains the raw cells and excluded seams as authority; it supplies no new cross-seam endpoint atlas. No merged-count experiment was run before the work stopped.

A literal one-parent-per-cell implementation would represent 644,699 parents before refinement. For scale only, **one chart per cell gives 644,699**, whereas **307 per cell gives 197,922,593**. These are **illustrations, neither predictions nor bounds** on the final chart count. No practical full-atlas runtime, memory, artifact-size or sector-count forecast was established. The unfinished closed-family continuation/compiler bridge and the remaining general boundary cases prevent promoting the local result to a benchmark.

2.3 Massive kite: a boundary is not a complete interval

The retained kite atlas has **9,840 original cells**. A genuine selected fourth boundary was admitted over all **ten replacement charts of cell 0**, with **40 strict checks at 128/256 bits**. This closes a boundary over three integration directions. It does not close the fourth interval, whose width, denominators and root-gap intersection with the upper face must join the complete original U, F, numerator and Jacobian inventory. In particular, $F^{-1-2\varepsilon}$ requires its own common-strip and symbolic subtraction treatment.

[1] Source/reuse reviews: `docs/reviews/no-deformation-positive-constant-endpoints.md`, `docs/reviews/no-deformation-recursive-endpoint-cover.md`, `docs/reviews/no-deformation-retained-margin-scale.md`, and `docs/reviews/no-deformation-canonical-measure.md`.

A small native divided-difference probe established exact candidate width and root identities, including higher-degree controls. Positivity on every closed face, complete interval density, automatic coverage of all 9,840 cells and full Laurent compilation/integration remain absent. Consequently the kite has no complete no-contour benchmark here. The accepted local proofs and generic implementation improvements remain useful without asserting a readily available whole-integral result.[2]

[2]The claim-to-source index, immutable outcome identities and cancellation records are in `docs/reviews/no-contour-final-scope.md`.

3 Landau pinches: what deformation can and cannot remove

A zero of a causal denominator, a trapped contour, a singular dependence on kinematics, and a pole in dimensional regularisation are different statements. The Landau conditions locate candidate singular configurations; the physical sheet and numerator cancellations must also be respected. A pinch alone does not determine whether the integrated answer diverges or only a derivative is singular. This distinction is explicit in Collins, §II.1.4. Here $i0$ denotes the causal boundary value and ε the dimensional regulator; they are not the same limiting parameter.

3.1 Why a strength change cannot unpinch the present map

For the real chart used by the contour methods,

$$w_i = x_i(1-x_i), \quad v_i = w_i \partial_i F, \quad z_i = x_i - i\lambda(x) v_i.$$

At an interior stationary zero $F(x_*) = 0$, $\operatorname{grad} F(x_*) = 0$, the direction vanishes exactly. Every finite smooth choice of strength leaves $z(x_*) = x_*$. Changing the safety factor S, strength cap L or displacement scale R cannot move this point. The usual local causal argument,

$$\operatorname{Im} F(x - i\lambda v) = -\lambda \sum_i w_i (\partial_i F)^2 + O(\lambda^3),$$

also has a vanishing leading term there. This is the same gradient-deformation mechanism described by Borowka, Carter and Heinrich, §2.2.

The full Jacobian still matters:

$$J_{ij} = \delta_{ij} - i\lambda \partial_j v_i - i v_i \partial_j \lambda.$$

Although $v(x_*) = 0$, its derivative need not vanish; the determinant is not generally one. Freezing $\operatorname{grad} \lambda$, dropping the determinant or making λ singular at the point would not preserve the admitted construction. A genuine causal pinch cannot be removed by an allowed nonsingular contour homotopy. A different representation can instead expose its regulated local contribution.

These statements concern an interior first-type configuration in a chart where U is nonzero; the gradient is with respect to the independent chart coordinates. Zeros of face weights, singular coordinate maps, $U = 0$, and intersections of several factors require separate analysis. Failure of this particular gradient direction, or a high-variance sample, is not by itself a complete Landau classification.

3.2 Detection: local evidence is not a domain-wide search

The current dynamic prescription does not automatically locate all stationary zeros. It retains F and $A = \sum_i w_i (\partial_i F)^2$. After certain terminal nonfinite or precision failures, a cold diagnostic evaluates this retained context with native certified balls. It reports a stationary-zero context only when both enclosures are exactly zero, rather than merely containing zero. This diagnostic also operates with optional causal validation off; it is not a hidden per-sample validation cost. It does not identify a unique failing density branch. A successfully evaluated density, including an exact numerator cancellation, does not trigger a rejection simply because its unreduced F has a stationary zero.

In the strict real interior, $w_i > 0$, so $A = 0$ is equivalent to $\operatorname{grad} F = 0$. On a face, vanishing weights hide normal derivatives: the same assertion then has only its weighted/tangential meaning. A finite pilot or production sample can miss an isolated point or a lower-dimensional locus. Small residuals, large weights and repeated precision rescue are useful candidates for investigation, not proof that a stationary zero exists or that all such zeros were found.

Nor is a collapsing strength a reliable warning. With constant positive U, consider $F_0 = \left(x - \frac{1}{2}\right)^2$ and $F_1 = 1 + \left(x - \frac{1}{2}\right)^2$. Their derivative envelopes and dynamic strengths are identical, although only the first has a stationary zero. At $x = \frac{1}{2}$, the polynomial construction gives $r = L$ and $\lambda = SL$: the displacement vanishes because $v = 0$, not because λ vanishes. Sign-aware smoothing may alter the limiting strength, but still does not distinguish the two cases. Successful radius solving therefore cannot certify absence of a pinch.

A proposed automatic preflight would instead solve the **critical system**

$$F(x;k) = 0, \quad \partial_i F(x;k) = 0, \quad 0 < x_i < 1$$

in the independent projective/chart coordinates at the requested kinematics k, retaining nonvanishing U and all original factor/domain conditions. Native derivatives and algebraic elimination could locate candidates; certified real-root isolation or subdivision must then establish existence and coverage. For a nonsingular Hessian, certified solution of $\operatorname{grad} F = 0$ still needs a separate exact check of $F = 0$ at that algebraic point. A ball containing zero is not that

equality check. Singular solutions and critical manifolds require additional algebraic/stratified treatment; ordinary Newton iteration is not a complete solver. Face strata require their own restricted equations.

Algebraic approaches to Landau loci are developed in Mizera and Telen, Landau Discriminants and Fevola, Mizera and Telen, Landau Singularities Revisited. Their qualifications about boundary/scaling strata matter: the displayed interior system is a useful restricted problem, not a complete classification of every graph singularity. The proposed preflight would have to distinguish **certified absent**, **certified loci found**, and **unresolved**. Evidence must be renewed on kinematic rebinding unless a parameter chamber proof applies. This solver orchestration is not currently implemented.

3.3 Local power counting does not imply a universal epsilon pole

An elementary local calculation illustrates the distinction. Suppose the critical point is isolated, the numerator and U are regular and nonzero, and the quadratic form is positive definite in m real variables. After a linear change of variables, the leading model is

$$F = \tau + |y|^2 + \text{higher orders}, \quad (F - i0)^{-\beta(\varepsilon)}.$$

Here τ measures distance from the threshold. For a scalar graph with ℓ loops and total propagator power N_ν, the standard parametric exponent is $\beta = N_\nu - \ell\frac{D}{2}$, $D = 4 - 2\varepsilon$; see the parametric representation, eq. (2). The following model identity follows directly by a Schwinger parameter and a Gaussian integral:

$$\int_{\mathbb{R}^m} \frac{\mathrm{d}^m y}{(\tau + |y|^2)^\beta} = \pi^{\frac{m}{2}} \frac{\Gamma(\beta - \frac{m}{2})}{\Gamma(\beta)} \tau^{\frac{m}{2}-\beta}, \quad \operatorname{Re}\tau > 0, \quad \operatorname{Re}\beta > \frac{m}{2}.$$

With a local cutoff, analytic continuation of this expression describes the leading nonanalytic term, up to analytic terms and a smooth nonzero factor. Consequently the exponent $\alpha = \frac{m}{2} - \beta(0)$ determines a possible threshold power: positive noninteger α gives a finite branch term, while negative α can give a power divergence. At nonnegative integer $\alpha = k$, the generic nonanalytic term is $\tau^k \log(\tau - i0)$, including a logarithm for $k = 0$, provided its coefficient survives; for example $\frac{1}{\Gamma(\beta)}$ can vanish. Apparent poles in the model's Gamma factor can cancel against the analytic part of the cutoff integral; they are not automatically poles of the complete regulated amplitude.

For example, $\int_{\mathbb{R}} \mathrm{d}\frac{y}{\tau+y^2} = \frac{\pi}{\sqrt{\tau}}$ for $\tau > 0$ has a threshold power singularity without any dimensional regulator. Conversely, numerator zeros can soften or cancel a singular term. A nonvanishing angular average of a leading numerator of degree σ shifts the local scaling to $\frac{m+\sigma}{2} - \beta$. An indefinite or rank-deficient Hessian, a critical manifold or a boundary pinch changes the analysis; replacing m by the Hessian rank without treating the remaining directions is unjustified.

A possible dimensional pole can be seen directly at $\tau = 0$. The radial power is $r^{m-1-2\beta(\varepsilon)}$. In the marginal case $\beta(\varepsilon) = \frac{m}{2} + b\varepsilon$, $b = \beta'(0) \neq 0$, a constant smooth term gives, apart from its angular coefficient,

$$\int_0^{r_0} r^{-1-2b\varepsilon}\, \mathrm{d}r = \frac{r_0^{-2b\varepsilon}}{-2b\varepsilon}, \quad \operatorname{Re}(b\varepsilon) < 0.$$

Continuation therefore exposes a possible simple ε pole. Its residue must still be combined with all sectors, numerators and prefactors. This explicit marginal mechanism is distinct from a general threshold power or logarithm at nonzero τ.

3.4 A possible resolution and subtraction route

Conceptually, one could identify the singular locus, split cells so that it lies on chart boundaries, and apply checked blowups or ramified maps until the complete density has the form

$$\prod_j t_j^{a_j + b_j\varepsilon + c_j\eta} h(t; \varepsilon, \eta).$$

The regular remainder must be controlled on every closed face. This requires all numerator, U, F, transported denominator and Jacobian factors, exact coverage of the original domain, and the original causal phase. It is not enough to factor F alone. Standard sector-decomposition termination results already assume that the input polynomials have no interior zeros; they do not automatically resolve a physical interior pinch (Bogner and Weinzierl, §3).

For one resolved variable, let $\rho = a + b\varepsilon + c\eta$ and $T_{K-1}h(t) = \sum_{k=0}^{K-1} h^{(k)}(0)\frac{t^k}{k!}$. The exact subtraction identity is

$$\int_0^1 t^\rho h(t)\, \mathrm{d}t = \sum_{k=0}^{K-1} \frac{h^{(k)}(0)}{k!(\rho + k + 1)} + \int_0^1 t^\rho (h(t) - T_{K-1}h(t))\, \mathrm{d}t.$$

It is first justified in a common convergence domain and then continued meromorphically; choose K so the remainder is locally integrable in the target regulator domain. The denominators display possible regulator poles. A pole contributes

only if its coefficient survives the complete sector and prefactor sum. If dimensional regularisation does not change a divergent direction ($b = 0$), an additional analytic regulator η, with its branch and removal prescription fixed, may be needed. Multiple-regulator subtraction is an established technique, illustrated by pySecDec, §4.8. It does not guarantee that every proposed resolution has a common starting domain. The limits $\tau \to 0, \varepsilon \to 0$ and $\eta \to 0$ must not be silently interchanged.

3.5 Targeted local blow-ups while retaining the contour

A hybrid route can avoid constructing a global sign decomposition of F. The proposed target is the usually smaller set $F = \operatorname{grad} F = 0$, not every regular point on the threshold hypersurface $F = 0$. Isolate and certify neighborhoods of the critical loci, resolve only those neighborhoods, and retain the contour on the complement. The published pinch-dissection examples of Gardi, Herzog, Jones and Ma provide a precedent for exposing a pinch by coordinate changes and domain splitting before sector decomposition. They do not establish a general fast algorithm for every topology. The construction below is our local derivation, not an implemented feature or a performance result attributed to that paper.

An explicit favorable case is an isolated interior stationary zero x_* with real nonsingular Hessian H, positive face-weight matrix $W_* = \operatorname{diag}(w_i(x_*))$, nonzero U and otherwise smooth density factors. Keep the original smooth causal map with finite $\lambda_* = \lambda(x_*) > 0$. Writing $x = x_* + y$ gives

$$z(x_* + y) = x_* + My + O(|y|^2), \quad M = I - i\lambda_* W_* H,$$

$$F(z(x_* + y)) = \frac{1}{2} y^T M^T H M y + O(|y|^3),$$

and hence

$$\operatorname{Im} F(z(x_* + y)) = -\lambda_* y^T H W_* H y + O(|y|^3).$$

Because H is nonsingular and W_* is positive definite, HW_*H is positive definite even when H is indefinite. The leading imaginary quadratic form is therefore strictly negative in every nonzero real direction. Moreover, W_*H is similar to a real symmetric matrix, so M is invertible. This argument uses the ordinary transpose, not the Hermitian transpose, as required by the analytic polynomial. Strength derivatives remain in the full Jacobian; they first affect higher terms here because $v(x_*) = 0$. The same local argument applies to an admitted positive fixed strength. Dynamic causal bounds help establish map admissibility; they do not perform the critical-locus search.

Choose a certified small cube $|y_i| \le \delta$ around the exact critical point. Split its 2^d orthants into d regions according to the largest $|y_i|$. In the region with largest coordinate p, use

$$y_p = \delta s_p r, \quad y_j = \delta s_j r t_j \quad (j \neq p),$$

$$0 \le r \le 1, \quad 0 \le t_j \le 1, \quad s_j \in \{-1, 1\}.$$

Its positive real volume factor is $\delta^d r^{d-1}$, with chart orientation accounted for. Boundaries shared by charts must be assigned consistently. Exact stationarity extracts

$$F(z(x_* + \delta r a(t))) = r^2 Q(r, t).$$

At $r = 0$, Q is its smooth derivative-defined extension, never a numerical $\frac{0}{0}$ quotient. Native symbolic derivatives supply its endpoint coefficients. The strict quadratic estimate and a certified higher-order remainder bound give $\operatorname{Im} Q \leftarrow c < 0$ on these closed angular charts for sufficiently small δ. The radial endpoint carries the pinch; the complex smooth factor has no local angular zero, including for a nonsingular saddle Hessian. A purely real blow-up has this simple unit property for a definite Hessian, but an indefinite one can leave a real angular zero cone. Retaining the contour is what removes that additional obstacle in the hybrid argument.

For a factor $(F - i0)^{-\beta(\varepsilon)}$, the radial measure becomes

$$r^{d-1-2\beta(\varepsilon)} h(r, t; \varepsilon).$$

Native symbolic subtraction can then expose any regulator pole as above, provided the complete regulated family and boundary derivatives are admitted. The original complex $\det J_z$ multiplies the radial volume factor exactly once. Numerators, positive factors and all derivatives of λ remain in h. Compact composed maps and finite symbolic endpoint coefficients would avoid expanding the entire substituted integrand.

For this particular isolated nondegenerate case, the explicit construction starts with **at most $d2^d$ local radial charts per point**: 8 in two dimensions, 24 in three, or 384 in six. This excludes complement charts, further factor resolution

and Laurent kernel splitting; it is not a bound for degenerate pinches or the final sector count. No number of such points has been established for the double-box benchmark. Its 644,699 sign cells do not count Landau points and do not predict the cost of a critical-locus search.

There are essential qualifications. Partition the pullback of the **same global contour**, or prove compatible homotopies across artificial interfaces; resetting the contour face weights independently in each new cube could freeze regular threshold zeros on new seams. Establish the partition identity in a common regulated convergence domain before continuing to the physical regulators. An algebraic critical center must remain exact/certified: approximate recentring leaves constant or linear terms and invalidates the exact r^2 extraction. Degenerate Hessians, nonisolated or intersecting loci, collisions with existing faces and vanishing U need further stratification or resolution. Near a threshold, a nonzero unfolding parameter also changes the radial form and may need its own parameter-dependent treatment. Proving that all relevant loci have been covered can still be expensive. Thus this hybrid plausibly avoids the specific global-GCAD cell explosion in favorable cases; it has no universal small-complexity or improved-variance guarantee.

3.6 Present implementation boundary

FastSecDec builds the weighted gradient and all map derivatives with native Symbolica operations. Its retained causal-context diagnostic can certify a stationary zero when reporting an unresolved deformation. It does not infer a unique failing density branch or a complete Landau classification, and a natively cancelled finite density remains allowed. Existing subtraction acts on admitted resolved charts; there is no automatic general Landau-pinch resolver or automatic extraction of new pinch-induced epsilon poles.

The splitting, resolution and auxiliary-regulator procedure above is a theoretical direction, not an implementation or a benchmark result in this report. At exact exceptional kinematics, the required result may be a kinematic limiting value or a regulated distribution rather than a finite ordinary integral. An unresolved proof or numerical failure cannot be replaced by zero.

4 Generic CAD and critical loci: exact scratch experiments

4.1 A generic decomposition is not a complete stratification

Full CAD can represent lower-dimensional solution cells. The present generic CAD interface does not return them as cells. Vitaly's original implementation recurses between roots and accepts strict inequalities; its API states that the original algorithm's separate boundary list is not computed. The Rust port has the same open-cell contract, with retained exceptional polynomials and independently checked generic coverage. This is a documented limitation for equality-locus queries, not evidence of a porting regression. Wolfram's documentation likewise distinguishes generic decomposition, with hypersurfaces containing omitted solutions, from full cylindrical decomposition. Knowing a hypersurface that contains exceptions is not the same as resolving all its solution strata.

A native executable control uses $0 < x < 1$, $F = (2x-1)^2$ and $F' = 8x - 4$. symGCAD returns and verifies two open cells, $\left(0, \frac{1}{2}\right)$ and $\left(\frac{1}{2}, 1\right)$, with sign vectors $(+,-)$ and $(+,+)$. Nevertheless, exact substitution gives $F\left(\frac{1}{2}\right) = F'\left(\frac{1}{2}\right) = 0$. The point survives in the root-boundary description; there is no returned zero-dimensional cell identifying it as critical. An equality constraint is explicitly rejected by this API. Thus a completed generic solve and the absence of zero-sign cells cannot certify absence of stationary zeros. A separate executable run of Vitaly's original code on $F > 0$ returns the same two raw intervals; its optional merge joins them into $(0, 1)$, absorbing the zero seam as its documented semantics allow. An algebraic zero seam must not be forgotten merely because signs agree on either side. This original-code probe took 7.8 ms.

For integration, this contract is usable if the partition identity holds in a common convergent regulated family and its endpoint terms are continued with the correct causal phases. The zero set need not be a separately sampled integration cell, but its singular neighbourhood cannot be discarded. Measure zero alone is insufficient after taking a distributional boundary value. The prior closed-face and analytic-continuation obligations therefore remain essential; these probes do not supply a missing general resolver. For example, splitting the following integral into its two open halves, initially for $\mathrm{Re}\,\varepsilon < 0$, gives

$$\int_{-1}^{1} (x - i0)^{-1-\varepsilon}\,\mathrm{d}x = \frac{1 - e^{i\pi\varepsilon}}{-\varepsilon} \longrightarrow i\pi \quad (\varepsilon \to 0).$$

Correct continuation retains the imaginary boundary contribution without a separate point integral. Dropping the phase or replacing the boundary value with an ordinary divergent real integral would lose it.

4.2 How native algebra and GCAD can cooperate

A targeted critical-locus analysis can solve the equations first, then use GCAD for the remaining inequalities. Native Symbolica differentiation, Groebner bases and algebraic solution facilities can reduce the critical ideal. On a positive projective chart, adjoining $h \prod_i x_i - 1 = 0$ excludes coordinate boundaries from this interior problem. A unit ideal proves absence. A non-unit ideal requires real-domain analysis; it is not evidence that a physical solution exists.

For an exact parametrized family, substitute into every original equation, prove the reverse implication under explicit nonzero guards, and certify the free-parameter domain and U with symGCAD. All denominator-zero and exceptional kinematic cases remain separate obligations. Where no parametrization is available, a complete equality/section solver or certified recursion on exceptional strata is needed. Calling generic CAD again in the same ambient space does not solve that problem.

Another sufficient absence test is an exact nonvanishing certificate for $P = F^2 + \sum_i (\partial_i F)^2$ on the real domain. The native domain-certificate API verifies such a certificate for the control $F = (2x-1)^2 + 1$ and correctly finds none for the stationary-zero control above. Failure to obtain this certificate is inconclusive. Neither a numerical minimum nor a ball enclosing zero establishes an exact equality.

4.3 Existing scalar points and threshold controls

The following scratch runs reuse native exported graph polynomials and exact kinematics. They concern the **strict positive projective interior**; boundary Landau configurations and kinematic chambers were not classified. The last Feynman parameter is fixed to one. Homogeneity and Euler's identity connect the chart equations with the full homogeneous gradient on $F = 0$.

Case	Exact point	Interior critical system
Massive bubble	$m^2 = 3, s = 16$	Absent: saturated ideal is (1)
Massive triangle	$m^2 = 1, s = 5$	Absent: saturated ideal is (1)
Massive box	$m^2 = 1, s = 5, t = -1$	Absent: saturated ideal is (1)
Massless sunrise	$s = 1$	Absent: saturated ideal is (1)
Massive sunrise	$m^2 = 1, s = 16$	Absent: saturated ideal is (1)
Massive kite	$m^2 = 1, s = 16$	Absent: ideal contains a strictly positive polynomial
Bubble threshold control	$m^2 = 3, s = 12$	Found: $(x_0, x_1) = (1, 1), U = 2$
Sunrise threshold control	$m^2 = 1, s = 9$	Found: $(x_0, x_1, x_2) = (1, 1, 1), U = 3$
Three-loop crown	$s = 1, t = -\frac{1}{5}$	Found: four-dimensional projective family

The triangle has two massless external legs; the box has massless external legs and $u = -4$. The kite has five equal massive propagators. For the kite, native grevlex reduction gives an ideal member proportional to $1 + x_0 + x_1 + x_2 + x_3 + 16x_0x_3$, strictly positive in this domain. Its complex critical variety need not be empty. A second native calculation verifies an explicit linear-polynomial combination of the five homogeneous derivatives equal to this positive polynomial, independently checking the obstruction. The final grevlex reduction took 4.22 ms and this identity took 8.23 ms; an earlier lex-order attempt was stopped after 2.50 minutes. The other scalar native calculations took less than 9 ms each. These are small diagnostic calculations, not timings for integral generation or a generic critical-locus solver.

The threshold controls also show why "only three loops can have a pinch" is too broad. It holds neither for arbitrary masses nor for exceptional external kinematics. Among these tested benchmark points the crown is the positive case; the claim does not cover the entire symGCAD corpus or the double box.

4.4 The crown: a critical manifold with three normal directions

The massless crown has **three loops and eight propagators**. No crown fixture was identified in the searched symGCAD benchmark inventory, so the scratch input transcribes the polynomial in Gardi, Herzog, Jones and Ma, §3.3. It is also the three-loop non-planar example in Olsson and Stone, §4.1.6. With zero-based parameter labels, set $x_1 = 1$ and introduce the invertible polynomial chart

$$a = x_0, \quad b = x_3, \quad c = x_5, \quad d = x_7, \quad x_2 = ab + u, \quad x_4 = ac + v, \quad x_6 = ad + w.$$

Native substitution and matrix arithmetic verify exactly

$$F = (s + t)duv - tcuw - sbvw, \quad \det H_\perp = 2st(s + t)bcd.$$

For $st(s + t) \neq 0$ and $b, c, d > 0$, the normal-gradient equations therefore force $u = v = w = 0$. Substitution makes F and all eight original derivatives zero. This proves conditional completeness of the positive interior family, not merely the existence of a candidate point. It has four free positive projective coordinates, and

$$U|_{u=v=w=0} = (a + 1)^3(bc + bd + cd + bcd) > 0.$$

GCAD then verifies the reduced four-parameter domain as one positive-U cell; changing its claimed sign makes verification fail. This call certifies the reduced inequalities, while the exact algebra locates the critical locus.

At $s = 1, t = -\frac{1}{5}$ and all $x_i = 1$, the native calculation gives $U = 32$ and Hessian rank three. The full seven-dimensional projective Hessian is degenerate; the previous isolated-point construction does not apply directly. The natural extension is a transverse, Morse–Bott-type treatment: retain the four tangent parameters and resolve the three normal directions, checking uniform bounds and the boundary/exceptional strata. For a regular local normal patch the earlier orthant/wedge count would start at $3 \times 2^3 = 24$; this is not a global sector-count prediction or the paper's distinct 24-ordering dissection. No crown integrand was generated or integrated here.

The entire crown scratch process took 0.0432 s and 12.7 MiB peak RSS; reduced GCAD solving and verification took 7.36 ms and 2.31 ms. It used existing warm native libraries, not a fresh latest-revision build. Scalar algebra uses Symbolica `ae0c82b`; the API control uses symGCAD `3baffa7` and the crown uses `bef2a9e`. The local `36bfd9e` API/source contract was separately audited. Exact revisions, source hashes, commands, independent review and limitations are recorded in `docs/reviews/gcad-critical-loci.md`. No production algorithm, dependency revision or numerical benchmark row changed.

5 Prospect: certified expansion by regions

The opportunity is a certified asymptotic expansion that also captures interior cancellation regions. A fast critical-locus solver could make its geometric preparation practical. It would not alone prove completeness. The desired output is a possibly redundant expansion to a requested order, with certified coverage and remainder control. A minimal nonzero-region list is a separate problem. The preceding scratch calculations are the only new experimental evidence for this research proposal.

5.1 Position relative to existing methods

The geometric method of regions associates scalings with lower faces of Newton polytopes, following Pak and Smirnov's geometric approach. The asy2 work of Jantzen and the Smirnovs already automates the exposure of potential and Glauber regions in its examples. Heinrich et al., Expansion by regions with pySecDec describes its numerical implementation and difficulties from cancellations in Minkowski kinematics. A region hidden by such a cancellation may become a usual facet region after a suitable change of variables. "Interior" here refers to the integration domain; it is not an invariant classification by location inside the original exponent polytope. The crown dissection in Gardi, Herzog, Jones and Ma, Dissecting polytopes provides an explicit example.

Gardi's Hidden Region Finder (HRF), September 2026 relates cancellation loci to local dissection and asymptotic scaling. Its §2.4.3 proposes a scaling-first search over exposed faces and positive Landau solutions, including boundary strata, as a theoretical alternative rather than the procedure used for its negative audits. The paper develops massless, unit-numerator examples; massive propagators and numerator-dependent contribution analysis are stated extensions.

Correia, Giroux and Mizera's SOFIA addresses a different problem: candidate kinematic singularities and symbol alphabets via Baikov representations and Fubini reduction. Its future directions distinguish singularity locations from local exponents and physical sheets. A certified pinch-manifold and density analysis could complement these outputs.

The homogeneous conditions $F = \operatorname{grad} F = 0$ with $U \neq 0$ concern finite-momentum, first-type candidates. Second-type singularities involve loop momenta at infinity and require additional strata. The critical equations for the Lee–Pomeransky polynomial $G = U + F$ have a different interpretation; see Fevola, Mizera and Telen, Landau Singularities Revisited. A real-domain calculation for a specified causal limit does not classify singularities on every complex sheet.

5.2 The limiting family contains more information than its critical set

Let $\rho \to 0^+$ specify the kinematic limit. As a local illustration,

$$F_2 = (x-a)^2 + \rho, \quad F_4 = (x-a)^4 + \rho, \quad x - a \sim \rho^{\frac{1}{2}} \ \text{ or } \ \rho^{\frac{1}{4}},$$

respectively. Neither polynomial has a real stationary zero for $\rho > 0$. Their limiting critical sets coincide, while their normal scalings differ. Thus a solve at one finite kinematic point, or just the zero set at $\rho = 0$, cannot determine the expansion. Vanishing orders, normal directions and all competing orders in ρ matter. The numerator, Jacobian and powers of U also contribute to power counting.

A proposed construction would combine the following obligations:

1. Fix an exact algebraic family, its physical chamber, causal sheet and requested asymptotic order. Cover projective boundaries and infinity, including relevant $U = 0$ strata and their intersections.
2. Reduce candidate leading equations with native Symbolica and use certified real solving on their admissible domains. Include scaling-selected leading polynomials: the full unscaled F need not itself be stationary on an asymptotic cancellation locus. Ordinary boundary regions remain necessary.
3. Construct exact local charts around each accepted component. Retain multiplicities and tangent parameters; resolve degenerate normal geometry, collisions and nested strata until the family admits controlled local expansions. Record branches, Jacobians, chart coverage and transitions.
4. Extract scalings in the resolved coordinates and compose them back to the original variables. Include the regular complement. Use a partition, or explicit overlap terms when expanded regions are extended beyond their original domains, so contributions are not counted twice.
5. Bound the integrated remainder in a common convergent regulator domain, then justify the required analytic continuation. Retain factors such as $\rho^{a+b\varepsilon}$ until the Laurent expansion; interchanging asymptotic and regulator limits without justification can lose logarithms.

This is a proposed certificate contract, not an existing general theorem or implementation in FastSecDec. Complete equality-aware CAD can reduce work on irrelevant ambient cells, as studied in England, Bradford and Davenport, CAD with equational constraints and Wilson et al., cylindrical algebraic sub-decompositions. Degeneracy, dependent equations and vanishing projection polynomials require their hypotheses or a complete fallback. The current symGCAD open-cell interface is not that equality solver.

5.3 Where our present work could improve the practical method

The promising comparison is the cost of obtaining a verified regional description, including failures and verification. It is not the cost of a critical-point probe versus somebody else's complete integral calculation. The following opportunities use actual native capabilities, but their benefits for region analysis remain to be measured.

Opportunity	Present evidence	Potential gain and remaining work
Target the critical set	Exact scalar exclusions; crown reduction to four free tangent parameters.	Avoid partitioning every regular threshold sheet. General component discovery and completeness are still needed.
Reject impossible candidates early	Native nonvanishing certificates and an explicit positive kite gradient identity.	Filter inadmissible real loci before expensive chart construction. A failed certificate remains unresolved.
Reduce projection cost	symGCAD certified positivity, linear/quadratic reductions and positive-root covers.	Simplify residual domain checks and massive quadratic structures, while preserving zero seams and exceptional cases.
Choose and reuse algebra	Bounded variable-order search, exact operation/proof caches and invertible factorization preconditioning.	Share work across related faces and branches. Heuristics may prioritize work, never silently remove candidates.
Keep the whole density	Native numerator support, symbolic derivatives, composed root/map evaluators and endpoint machinery.	Compute numerator-dependent orders and test regional coefficients. General family resolution and remainder certificates remain missing.

The algorithm inventory is pinned to symGCAD's implementation checkpoint. Some techniques are optional or experimental. Its matched generic-CAD study reports a 6.82-fold geometric-mean reduction in matched algebra-phase time against the original implementation. That result is neither an end-to-end region-analysis comparison nor a guarantee for every problem; the checkpoint still records unresolved fixed and parametric cases.

The most concrete extension beyond the HRF study's stated scope would be **massive families with algebraic critical branches and realistic numerators**. Our exact root/domain infrastructure and symbolic density machinery make this a plausible target. They do not establish that the difficult general algebraic resolver is complete, or that our approach is already faster than HRF or SOFIA. Shared geometry could be reused across numerator terms and Laurent outputs only with matching kinematic and branch guards.

5.4 Surplus candidates, exact cancellation and the completeness test

Projection artifacts or solutions outside the physical domain can be excluded algebraically. Other candidates describe equivalent charts, scaleless contributions, numerator cancellations or overlap terms. Each exclusion needs its corresponding exact argument and regulator prescription. A region that vanishes at leading order can still contribute at a later order. It is safer to retain a harmless redundant contribution than to discard it on the basis of a small numerical estimate.

Jantzen, Foundation and generalization of the expansion by regions derives regional identities with explicit covering/convergence assumptions and overlap terms. Cancellation of regulator poles does not prove completeness: an omitted region can be finite. Nor does agreement at a finite set of kinematic points establish that all orders or parameter strata were covered. Exact coefficient cancellations can require integral identities, making a minimal nonzero-region list harder to obtain than a complete redundant expansion.

A useful first comparison would reproduce ordinary Euclidean regions, the massive threshold controls and the crown's known hidden scaling, followed by HRF's moving and boundary examples. Record discovery, local resolution, verification, cell/chart counts and aggregate memory separately; compare exact scalings and coefficient expansions before comparing speed. Include negative cases and deliberately incomplete searches. The central acceptance criterion must be **certified coverage plus a controlled remainder**, not just finding more candidate regions or observing their cancellations.

6 Numerical compatibility

The original campaign audit compares 126 accepted native production estimates. The tests use the full real–imaginary covariance, retained before rounding for display. For two independent runs, the difference uses $C_1 + C_2$. The diagnostic distance is $\sqrt{\Delta^T C^{-1} \Delta}$. It is an estimated-covariance diagnostic, not a calibrated probability or a proof of correctness. Distinct recorded seeds support the independence assumption; the audit does not manufacture empirical cross-run covariance. Every available complex Laurent block is checked; the summary below concerns ε^0.

Case	Runs	Largest mutual distance	Largest reference distance
Massive bubble	18	2.39	2.39
Massive triangle	22	2.77	2.58
Massive box	22	2.46	2.17
Massless sunrise	22	3.24	3.08
Massive-threshold kite	18	3.25	—
Scalar D05 double box	12	2.72	—
Physical D05 double box	12	2.74	—

The checks include comparisons between samplers, prescriptions and work/time arms where results are available. The largest nonsingular mutual distance over all orders is 3.78; the largest reference distance is 3.08. These many comparisons are correlated with one another and are not independent repetitions of a significance test. Later campaign closures and instrumented reruns require the same audit. The current all-order maximum comes from the physical double box at ε^{-1}, involving its noisy dynamic QMC estimate. The finite-coefficient maxima are listed separately above; no uncertainty has been enlarged to obtain agreement. A clean capped secondary-work estimate can be included with its actual count and unfinished planned allocation; its covariance still comes from complete native batches. An overlapping partial prefix is omitted when its completed continuation is used.

6.1 Independent references and numerical limits

The added massive-sunrise block has a separate audit of its two fresh timed estimates. All 36 full-Laurent component comparisons with the previously validated fixed, dynamic and pySecDec controls pass the existing eight-combined-SE criterion without an added floor; the largest difference is 2.162 combined SE. Both complete native covariance matrices and the radial norm uncertainties are checked independently. These comparisons are recorded in `no-contour-massive-sunrise-overview.json`; they do not change the original campaign counts above. The subsequently completed six contour timings and their separate native joint audit are recorded in the overview details and `massive-sunrise-contour-overview.json`.

The bubble's finite reference is $2 - 3\frac{\log(3)}{2} + i\frac{\pi}{2}$. Native HEPKit/OneLOop expression evaluation supplies the triangle and box references at exactly the listed invariants and masses. Their measure includes the explicit inverse r_Γ factor. The massless sunrise uses Symbolica's Laurent expansion of the composed analytic bubble identity, including its causal phase. Fresh native reference fixture checks passed for the triangle, box and sunrise; the historical kite reference is deliberately excluded because its kinematics and masses differ from this campaign.

There is currently no independently established integral reference in this campaign for the five-massive kite at $p^2 = 16$ or either D05 double box. Agreement between implementations there is a useful consistency check and does not substitute for an analytic reference. The scalar and physical-numerator double boxes are distinct integrals despite sharing denominator geometry.

Bubble QMC uncertainties can be smaller than binary64 rounding effects. Singular or rounding-sensitive covariance tests are explicitly retained as unresolved diagnostics and excluded from the maxima above. Covariance is never padded or replaced by a diagonal matrix to obtain agreement. The rounding-warning scale is a heuristic, not an error bound; the reported sampling SE excludes shared numerical bias. Rounded reference arithmetic likewise has no certified enclosure and is not silently counted as additional stochastic error.

The compact reference associations and comparison results are committed as `benchmark-compatibility-references.json` and `benchmark-compatibility-results.json`. The native `benchmark_compatibility` example reuses Numerica's exact conversion of recorded binary64 entries and its matrix determinant/solve. It checks the native saved result, input-card association and distinct seeds without generating or sampling. Full covariance and raw estimates remain in the benchmark data; private audit receipts preserve every pair and coefficient rather than only these maxima.

The final double-box tuning extension has its own 26-estimate audit in `double-box-dynamic-work-overview.json`. Its 234 complex-block comparisons include six complete equal-work QMC controls as well as the timed baselines and confirmations. The largest distances are 3.169 (scalar finite), 3.805 (physical pole) and 3.280 (physical finite), without non-positive covariance or rounding-sensitivity flags. This does not replace the original broader campaign audit or discard its earlier high-variance and roundoff diagnostics.

Threshold regularisation

FastSecDec · contour deformation and threshold decomposition

Benchmark campaign closure · 12 October 2026

7 Deformation styles and formulae

7.1 Common geometry and fixed strength

After sector mapping, separate the real endpoint monomials from the smooth density. Let $F(x)$ denote the designated residual causal polynomial and let $U(x)$ denote each retained positive residual factor. The definition of F includes its physical scale: rescaling F also rescales this deformation. For a sampled sector with real masses, real kinematics and $x \in [0,1]^n$ ($n \geq 1$), define

$$w_i = x_i(1-x_i), \quad v_i = w_i \partial_i F, \quad z_{i(x)} = x_i - i\lambda(x) v_{i(x)}.$$

Exact zero-dimensional contributions bypass sampling. The coordinate faces remain fixed in their normal direction. With $A = \sum_i w_i (\partial_i F)^2 \geq 0$, the ray obeys

$$\operatorname{Im} F(x - itv) = -tA + O(t^3).$$

Thus a sufficiently small positive strength moves the causal denominator to the lower half-plane. The scalar λ is a strength, not the norm of the displacement; no undocumented normalization by F or its gradient is used. Since cube coordinates are dimensionless, $[\lambda] = [F]^{-1}$; the same units apply to L. The displacement limit R is in parameter-space units.

Fixed mode takes a finite runtime value $\lambda(x) = \lambda_0 > 0$. Its Jacobian is

$$J_{ij} = \delta_{ij} - i\lambda_0 \partial_j v_i, \quad \partial_j v_i = \delta_{ij}(1-2x_i)\partial_i F + w_i \partial_j \partial_i F.$$

The complex, oriented determinant $\det J$ multiplies the density. A zero determinant alone is not a pole crossing. Neither its absolute value nor a frozen-gradient approximation gives the same contour integral.

For endpoint exponents $a_i + b_i \varepsilon$, the real endpoint monomials are retained and the smooth density receives

$$\det J \prod_i \rho_i^{a_i + b_i \varepsilon}, \quad \rho_i = \frac{z_i}{x_i} = 1 - i\lambda(x)(1-x_i)\partial_i F.$$

The last expression defines ρ_i even at $x_i = 0$. Deform the numerator and residual factors as well. Apply this pullback **before** Taylor/IBP subtraction and Laurent construction. Boundary terms restrict the same full-sector map; they do not construct a new, lower-dimensional deformation.

7.2 Branches and checking policy

Continue each factor separately. Endpoint ratios have real part one; positive residuals must keep the appropriate branch, and the causal logarithm has the lower-lip value $\log(F - i0) = \log(|F|) - i\pi$ when $F < 0$. Combining these continued logarithms into the principal logarithm of their product can change the sheet. A stationary point with $F \neq 0$ is allowed. A surviving singularity at $F = 0$ and $A = 0$ requires an unresolved-deformation diagnostic, not a fabricated zero; a natively cancelled denominator or finite smooth density remains finite.

Optional validation has three policies: **always** checks pilot and production, **pilot** checks separate pilot work and removes optional production checks, and **off**, now the default, removes both. Solver termination, precision rescue and nonfinite handling remain active. The new overview's sampling and convergence timings explicitly use **off**; separate validation probes are excluded. Fixed-mode failures identify the factor and point; they never silently shrink the requested strength. Finite pilots are safeguards, not global certificates. The gradient-map convention and subtraction ordering follow the established sector-decomposition approach; see pySecDec.

7.3 Dynamic strength: a smooth sufficient causal radius

Set $\lambda(x) = Sr(x)$ with $0 < S < 1$. The radius bounds the **whole homotopy** $x - itv(x)$ for $0 < t \le Sr(x)$. It is not the distance to the nearest complex root: a positive real ray root need not exist, and a root-modulus prescription can vanish at the very physical pole one must avoid.

In the auxiliary ray expansion, hold $v(x)$ fixed. For odd $k \ge 3$, define

$$T_k = D^k F(x)[v^{k-2}, \cdot, \cdot], \quad B_k = \frac{\sum_{i,j} w_i w_j (T_k)_{ij}^2}{(k!)^2}.$$

For every positive residual U, write the real even-ray expansion

$$\operatorname{Re} U(x - itv) = U(x) + \sum_{k \ge 2,\ \text{even}} d_{k(x)} t^k, \quad C_k = \frac{d_k^2}{U(x)^2}.$$

Native polynomial positivity admission supplies a strictly positive residual U on the closed cube; an arbitrary positive floor is not substituted. Let m_F and m_U count the structurally retained odd and even orders. With positive cap L and displacement scale R, solve

$$H(x, r) = \frac{r^2}{L^2} + r^2 \frac{\|v\|^2}{R^2} + m_F \sum_{k \ge 3,\ \text{odd}} B_k r^{2k-2} + \sum_U m_U \sum_{k \ge 2,\ \text{even}} C_k r^{2k} = 1.$$

All coefficients are nonnegative. The cap term makes the positive root unique and simple, with $r \le L$ and $rH_r \ge 2$ at the root. Both constraints are smooth; there is no hard minimum whose derivative would jump.

Why the bound is causal. Put $W = \operatorname{diag}(w_i)$ and $q = \sqrt{W}\operatorname{grad} F$, so $A = q^T q$ and $v = \sqrt{W} q$. The higher odd ray term satisfies

$$\left| D^k \frac{F[v^k]}{k!} \right| \le A \sqrt{B_k}, \quad \sum_k \sqrt{B_k} t^{k-1} \le \sqrt{m_F \sum_k B_k t^{2k-2}}.$$

Consequently higher imaginary terms cannot cancel the leading $-tA$ along the admitted homotopy when $A > 0$. The even-factor bound similarly keeps $\operatorname{Re} U(z) > 0$. The proof does not divide by A, so small gradients do not create an artificial singular bound. At $A = 0$, the stationary-point rules still apply.

7.4 Full local-strength derivatives

Dynamic strength changes the Jacobian:

$$J_{ij} = \delta_{ij} - i\big(\lambda \partial_j v_i + v_i \partial_j \lambda\big).$$

Implicit differentiation of $H(x, r(x)) = 1$ gives

$$\partial_j r = -\frac{\partial_j H}{\partial_r H}, \quad \partial_j \lambda = S \partial_j r.$$

Here $\partial_j H$ holds r fixed but differentiates all physical coefficient dependence, including v and the smooth constraints. Symbolic endpoint reduction differentiates the complete deformed density, including this identity and the determinant. Solver iteration choices are never differentiated. Every face retains the full-sector dimension, structural counts and cap parameters. Omitting the rank-one term $-iv(\operatorname{grad}\lambda)^T$ changes the integral.

The polynomial construction is an independently checked baseline. Its exact-arithmetic proof is distinct from numerical validation of a rounded implementation. Disabling optional checks removes their runtime cost, not solver failures or nonfinite-value diagnostics.

7.5 Sign-aware refinement of the dynamic bound

The baseline bounds beneficial as well as harmful higher terms. A smoother, tighter alternative bounds only terms that can oppose causality. Define the real symmetric matrix

$$R_k = \frac{(-1)^{\frac{k+1}{2}} L^{k-1}}{k!} \sqrt{W} T_k \sqrt{W}.$$

It need not be built with explicit square roots of W. Weighted trace and squared-norm identities provide the native polynomial quantities

$$a_k = \frac{\operatorname{tr}(R_k)}{n}, \quad b_k^2 = \frac{n-1}{n}\left(\operatorname{tr}(R_k^2) - \frac{\operatorname{tr}(R_k)^2}{n}\right).$$

For a symmetric matrix its largest eigenvalue is at most $a_k + \sqrt{b_k^2}$. Smooth both the spectral bound and its positive part with a fixed dimensionless $\delta = 10^{-3}$:

$$t_k = a_k + \sqrt{b_k^2 + \delta^2}, \quad p_{\delta(t)} = \frac{t + \sqrt{t^2 + \delta^2}}{2}, \quad \mu_k = p_{\delta(t_k)}.$$

For negative t, evaluate the exactly equivalent expression

$$p_{\delta(t)} = \frac{\delta^2}{2\left(\sqrt{t^2 + \delta^2} - t\right)}$$

to avoid cancellation. For each even U coefficient, use

$$\nu_{U,k} = p_{\delta\left(-d_k \frac{L^k}{U}\right)}.$$

With $u = \frac{r}{L}$, form

$$E_{F(u)} = \sum_{k \geq 3,\ \text{odd}} \mu_k u^{k-1}, \quad E_{U(u)} = \sum_{k \geq 2,\ \text{even}} \nu_{U,k} u^k.$$

Then solve the unique positive root

$$u^2 + \frac{L^2 u^2 \|v\|^2}{R^2} + E_{F(u)}^2 + \sum_U E_{U(u)}^2 = 1, \quad \lambda = SLu.$$

The nonnegative envelope coefficients retain monotonicity and smoothness. Their derivatives, including smoothing, enter the same implicit-jet machinery. A larger permitted displacement is a diagnostic; it does not establish lower variance or lower time to precision. The baseline and sign-aware construction must be compared with their full evaluation costs.

7.6 Sampling, diagnostics and interpretation

Korobov remains an outer real map: $y = T(q)$ followed by $z = z(y)$. Its total Jacobian is $\det J_{z(y)} \prod_i T_{i'}(q_i)$, with each factor applied once. No inverse Korobov derivative is introduced at an endpoint. Contour evaluation is deterministic and consumes no production random streams.

QMC uncertainties come from complete independently randomized lattices; democratic QMC retains covariance between sectors sharing a shift. Discrete Havana MC retains its frozen production proposal and uses complete batches. Both accumulate the full Laurent vector and its covariance. A partial allocation, a missing sector or an unvisited zero is not exact evidence.

The smooth sufficient-radius bounds above are FastSecDec's parameter-space construction. They are not attributed to pySecDec or to the LTD algorithm. Related work includes Winterhalder et al., *Targeting Multi-Loop Integrals with Neural Networks* on optimized contour choices, and Capatti et al., *Numerical Loop-Tree Duality: contour deformation and subtraction* on causal contour construction and physical multiloop tests. Their timings are not directly comparable FastSecDec speed measurements.

8 Physical $gg \to hh$: the 1000 GeV double box

The input is the existing individual **D05** s-channel double-box contribution, with incoming $++$ helicities, $\sqrt{s} = 1000$ GeV, $\cos(\theta) = \frac{4}{5}$, $m_H = 125$ GeV and $m_t = y_m = 172.5$ GeV. HEPKit supplies the native graph, contracted numerator, wavefunctions and kinematics. The colour contraction is unnormalized δ_{ab}; the measure is $\prod_l d^D \frac{k_l}{i\pi^{\frac{D}{2}}}$, $D = 4 - 2\varepsilon$, with multiplier one. This is not a complete diagram sum or a spin/colour averaged matrix element.

Both prescriptions use symbolic endpoint IBP. Fixed mode uses the symbolic Jacobian; dynamic mode uses contour-only dual evaluation. Both complete artifacts contain 30 sampled sectors and 30 exact records.

8.1 Complete generation

Contour	Workers	Minutes	Peak GB	Saved MB	Exact / JIT MB
Fixed	6	43.57	54.09	70.41	3.20 / 4.66
Dynamic	2 ⟶ 4	106.81	60.94	51.34	11.33 / 15.98

Both use serial generation, SymJIT O2, Horner zero and common-pair cap 1000. Dynamic time includes the initial two-worker attempt and its four-worker continuation; the 26.1-second restart gap is excluded. Initial native series widths are one and two, respectively. These are complete-campaign observations, with different worker counts; the matched Jacobian comparison is in section 4. GB and MB are decimal; RSS includes the owned parent and child processes.

No-contour status: the double-box threshold-decomposition comparison is not available yet. The new no-contour profiles below cover complete small inputs. The extended exact-point GCAD trial in section 3 now completes with independent verification; the physical density and endpoint-resolution bridge remains open.

8.2 Five minutes on 50 physical cores

Method	Contour	Finite coefficient	Joint variance	Relative SE
QMC	Fixed	$68.35 + 26.24i$	2.46049	2.142%
QMC	Dynamic	$69.45 + 24.11i$	7.8327	3.807%
Discrete MC	Fixed	$69.07 + 24.93i$	0.256372	0.690%
Discrete MC	Dynamic	$68.45 + 24.65i$	0.500102	0.972%

For the finite complex estimate $\widehat{I}_0$, the joint estimator variance is $V = \operatorname{tr} \Sigma_0$ and relative joint standard error is $\frac{\sqrt{V}}{|\widehat{I}_0|}$. This is not pointwise integrand variance. The reproducibility JSON retains both diagonal covariance entries and the off-diagonal entry, all Laurent components, exact offsets, work counts and precision-rescue diagnostics.

Method	Contour	Seconds	Peak GB	Sample µs avg / max
QMC	Fixed	302.32	1.94	3.73 / 4.02
QMC	Dynamic	302.37	2.02	16.07 / 28.97
Discrete MC	Fixed	303.53	4.13	23.99 / 43.54
Discrete MC	Dynamic	302.27	11.31	71.79 / 164.48

Sample costs are the average and maximum of per-sector mean integrand costs, including precision rescue and amortized setup/adaptation work; they are not maximum single-point latency. A separate fixed all-sector probe measured 10.44 µs average and 14.06 µs maximum sector mean on a smaller fixed point set.

Fixed uses $\lambda = 10^{-6}$; dynamic uses $S = .8$, $L = 10^{-6}$, $R = 1$, selected with separate tuning seeds. Dynamic/fixed variance ratios are 3.18 (QMC) and 1.95 (MC): fixed performs better in these runs. Each row is one independent production run; these ratios have no repeated-run confidence interval. The 300-second native integration budget includes worker setup and Havana adaptation. Loading, binding and causal Pilot16 precede that clock; optional production causal checks are disabled. Complete accepted work is drained after cancellation, explaining the modest overshoot. All rows retain full covariance and independently reserved production streams; incomplete work is not added. No independent numerical reference exists for this point. These frozen campaign executables precede the loader-cache update in section 4.

8.3 Scalar topology comparison: generation and sampling cost

The four native scalar inputs use numerator one and $D = 4 - 2\varepsilon$, through the finite term. All six constructions retain symbolic endpoint IBP, SymJIT O2, Horner zero, common-pair cap 1000 and initial native series width two. Generation uses one in-process caller and one compiler core; sampling uses one core per task, with at most two tasks running on distinct physical cores. These are separate from the 50-core double-box measurements.

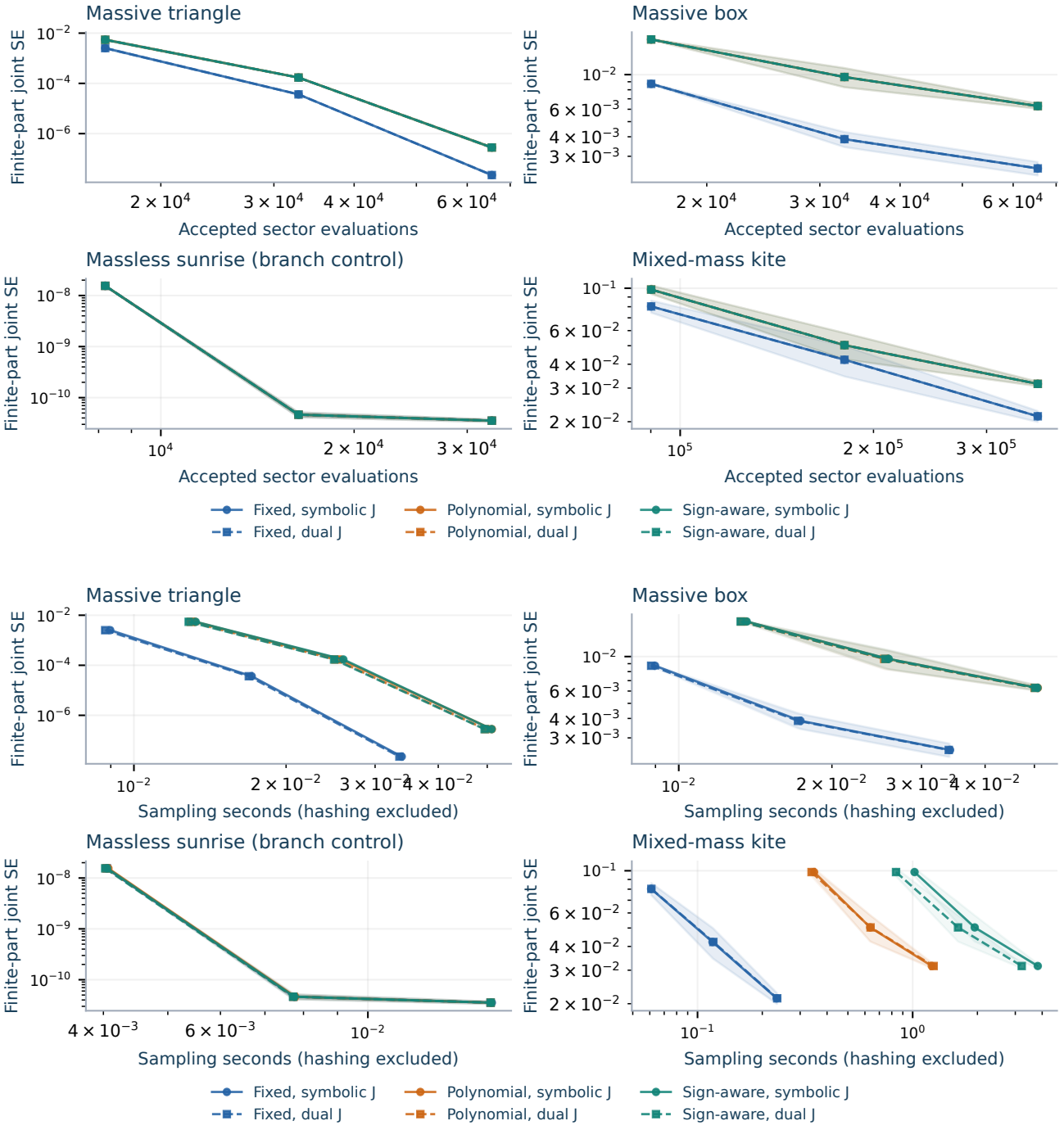
Case	Contour	J	Gen. s	Gen. MB	IR kB	Eval. µs avg/max	Run MB
Triangle	Fixed	S	0.022	381.26	2.202/3.216	0.248/0.281	16.13
	Fixed	D	0.022	450.16	1.710/3.402	0.241/0.271	16.05
	Polynomial	S	0.027	435.97	5.616/7.442	0.503/0.582	16.33
	Polynomial	D	0.028	454.82	4.470/7.785	0.483/0.549	16.44
	Sign-aware	S	0.025	458.05	5.616/7.442	0.503/0.581	16.45
	Sign-aware	D	0.028	406.97	4.470/7.785	0.480/0.543	16.24
Box	Fixed	S	0.032	471.89	3.212/5.166	0.222/0.228	16.10
	Fixed	D	0.031	476.60	2.492/5.035	0.218/0.222	16.02
	Polynomial	S	0.039	469.16	8.366/11.91	0.469/0.480	16.62
	Polynomial	D	0.043	467.39	6.626/12.97	0.460/0.471	16.39
	Sign-aware	S	0.038	466.14	8.366/11.91	0.469/0.481	16.39
	Sign-aware	D	0.043	467.23	6.626/12.97	0.461/0.479	16.66
Sunrise	Fixed	S	0.020	59.79†	0.846/0.975	0.160/0.171	15.80
	Fixed	D	0.019	59.78†	0.812/1.065	0.158/0.170	15.84
	Polynomial	S	0.021	59.80†	0.862/0.975	0.160/0.171	15.90
	Polynomial	D	0.021	59.79†	0.828/1.065	0.159/0.171	16.01
	Sign-aware	S	0.021	59.79†	0.862/0.975	0.159/0.171	15.88
	Sign-aware	D	0.021	59.82†	0.828/1.065	0.159/0.171	16.05
Kite	Fixed	S	0.358	21.48	40.56/85.37	0.328/0.351	21.18
	Fixed	D	0.269	18.88	44.62/100.26	0.329/0.369	20.33
	Polynomial	S	1.967	62.48	674.99/1082	3.051/4.733	58.08
	Polynomial	D	3.801	59.31	692.68/1240	3.134/4.946	57.33
	Sign-aware	S	4.022	114.91	1237/2016	10.20/19.15	92.34
	Sign-aware	D	12.30	105.32	1156/1877	8.558/14.94	81.16

S/D denotes symbolic/contour-only dual Jacobian. Gen. time includes native generation and evaluator compilation, excluding input/reference evaluation and serialization. IR gives exact / lowered JIT program sizes. Generation RSS covers the whole input/reference, generation, compilation and saving process; it therefore has a broader scope than the generation timer. Memory is sampled owned-process peak RSS in decimal MB; run peaks include loading, pilot and all three sampling epochs. Evaluator costs include precision rescues: average is the mean of sector means over the two final-epoch runs, maximum is the largest sector mean in either run. They exclude caller overhead and are not maximum single-point latencies. A dash denotes missing accepted evidence. † The original sunrise processes finished between RSS polls. These six values are kernel high-water measurements from separate identical generation-only repeats, retaining the original generation timings. GNU time's parent RSS was independently below 1.81 MB, excluding the inherited launcher-memory floor. All other generation RSS values are periodic samples; short peaks may be missed.

The independent references are native OneLOop expressions for the massive triangle and box, a Symbolica Laurent expansion of the gamma-function identity for the massless sunrise, and the published convergent small-momentum series for the kite. Points are $C_0(0,0,5;1,1,1)$, $D_0(0,0,0,0,5,-1;1,1,1,1)$, $p^2 = 1$ for the sunrise, and $p^2 = \frac{3}{1000}$, $m^2 = 1$ for the three-massive/two-massless kite. The sunrise probes causal branches and UV endpoint subtraction; it has no interior causal zero. References: OneLOop Fleischer, Smirnov and Tarasov, zero-threshold expansion. Native normalization and routing checks accompany the fixtures.

8.4 Scalar convergence at matched work and elapsed time

Each curve uses two run seeds and separate complete lattice epochs of 1024, 2048 and 4096 points, with eight shifts, Kuo33002 and Korobov3. The displayed line is the arithmetic mean of the two native joint standard errors; shading spans the two observations. No estimates or covariance matrices are pooled across lattice sizes, seeds or constructions. The band is not a confidence interval.



Sampling time excludes separately measured coordinate hashing and includes native worker-context creation, precision-cache warm-up and caller work. Each x-coordinate is the mean of the two epoch times. These short runs illustrate finite-work behavior; they do not establish an asymptotic convergence rate.

8.5 Scalar findings and limits of the comparison

All 24 generated owners and all 144 lattice epochs completed. All 72 symbolic/dual Jacobian pairs have identical actual coordinate and weight hashes, with full-vector means agreeing to 1.47×10^{-16} on the scale $\max(1, |I|)$. Every case admits the common fixed strength or dynamic cap 0.1; dynamic modes use $S = .8$, $R = 1$. This is a common-cap comparison, not an optimization of each prescription's variance. In particular, dynamic strength is at most .08, whereas fixed strength is .1.

Case	Contour	Mean V	$\frac{V}{V_F}$	Rel. SE %	Slope p	Time ms
Triangle	Fixed	5.020e-16	1.00	2.07e-06	8.39	33.87
	Polynomial	7.888e-14	157.13	2.6e-05	7.13	51.06
	Sign-aware	7.888e-14	157.13	2.6e-05	7.13	50.98
Box	Fixed	6.354e-06	1.00	0.636	0.90	34.14
	Polynomial	3.963e-05	6.24	1.59	0.71	50.77
	Sign-aware	3.963e-05	6.24	1.59	0.71	50.69
Sunrise	Fixed	1.260e-21	1.00	1.72e-09	4.39	15.36
	Polynomial	1.260e-21	1.00	1.72e-09	4.39	15.31
	Sign-aware	1.260e-21	1.00	1.72e-09	4.39	15.32
Kite	Fixed	4.579e-04	1.00	0.457	0.96	233.31
	Polynomial	9.969e-04	2.18	0.676	0.82	1227.83
	Sign-aware	9.969e-04	2.18	0.676	0.82	3815.49

The table uses symbolic J, with each entry a descriptive mean over the two runs at 4096 points × eight shifts per sector. V is native estimator variance, V_F the fixed result, and $p = \log_4\left(\frac{\mathrm{SE}_{1024}}{\mathrm{SE}_{4096}}\right)$ uses the two-seed mean standard errors. The large triangle slope is an observed short-range decrease, not a demonstrated scaling law. No new combined estimate is constructed.

Fixed strength wins at these settings. The polynomial dynamic variance is about 157 times larger for the triangle, 6.24 for the box and 2.18 for the kite at matched work. The dynamic evaluator also costs more per point. These results do not rule out better choices of the caps or a benefit on different physical inputs.

Two degeneracies explain overlapping curves. The triangle and box have quadratic residual F and linear U. There are no higher odd causal terms or even positive-factor terms to distinguish the two dynamic envelopes, so polynomial and sign-aware reduce to the same radius equation. For the massless sunrise, monomial extraction leaves constant residual F: all dual build records report zero surviving contour derivative slots, and all six estimates agree. It is a useful branch/subtraction control, not evidence that six active deformation algorithms have identical cost.

The kite distinguishes the full envelopes. Its sign-aware construction offers essentially the same variance as the polynomial bound at this admitted cap, at greater generation and sampling cost. Contour-only dual J reduces the sign-aware mean evaluator cost from 10.20 to 8.56 µs per sector-point, but generation rises from 4.02 to 12.30 s. It is therefore not a universal generation optimization. The complete timing and memory records include both constructions.

At the final lattice size, the largest finite-reference error norm divided by the native joint standard error is 1.88 across the suite. This descriptive check is consistent with the references; two run seeds cannot calibrate uncertainty coverage. There are no failed, unstable or cutoff-zero evaluations, no arbitrary-precision rescues, and no optional production causal checks. Double-double precision rescues remain enabled and are included in the measured cost. Full Laurent vectors, including the sunrise pole, are retained in the JSON.

The campaign took 90.23 s with 0.961 GB aggregate sampled RSS, including its coordinator and at most two active processes. The optimized example link took 143.04 s and the OneLOop reference leaf 4.83 s; neither is part of generation timing. The generation/reference/protocol controls and independent ecosystem review pass. The wider Phase B scientific acceptance programme remains unfinished.

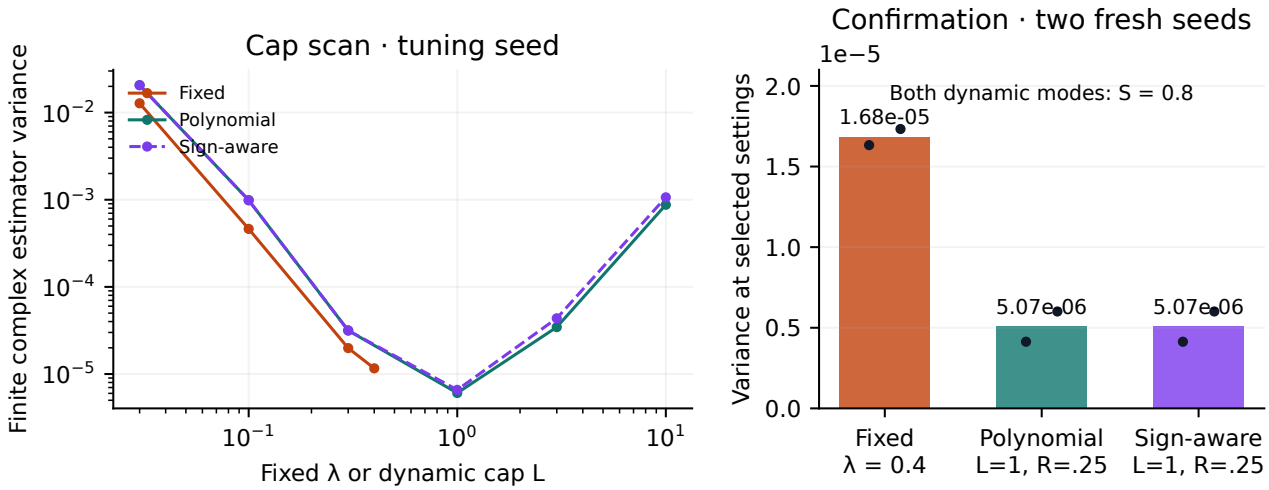
8.6 Cap tuning with the normal Horner setting

The follow-up uses the same connected scalar kite, now generated with the public **Horner-ten** default. All three rows retain symbolic endpoint IBP and a symbolic deformation Jacobian. Its 22 source charts give 11 numerical kernels. Changing L, R or S reuses these saved evaluators without regeneration or Horner optimization. SymJIT O2 and the earlier physical normalization remain unchanged.

Construction	Gen. s	Peak MB	Exact / JIT kB	Eval. µs avg/max
Fixed	.455	56.62	24.16 / 57.35	.274 / .297
Polynomial	2.525	68.31	190.59 / 409.78	1.742 / 2.404
Sign-aware	5.039	115.95	236.90 / 473.36	2.566 / 3.588

Generation is native symbolic construction plus evaluator compilation. Peak memory is GNU time's process high-water mark over input/reference, generation, compilation and saving. Sampling costs use the selected settings below, including precision rescues. The average is the mean of sector means over two confirmation runs; the maximum is the largest sector mean in either run. Horner ten reduces the polynomial/sign-aware lowered IR from the historical 1.08/2.02 MB to .41/.47 MB. Those older programs remain labelled Horner zero.

A predeclared tuning scan used $L = 0.03, 0.1, 0.3, 1, 3, 10$, with $S = .8$, $R = 1$, and a separately tuned fixed strength. Fixed .5, .6, .8, 1 failed the native causal pilot; their failures are retained rather than silently shrinking them. A second scan varied $R = .25, 4$ and $S = .5, .95$. Choices were frozen before two fresh confirmation seeds. Each arm uses 4096 points × eight shifts per sector, Kuo33002 and Korobov3. Actual point/weight hashes match across prescriptions within each seed, and differ between seeds. All estimates retain native complex covariance and exact offsets; separate arms and seeds are never pooled.



The left panel is one tuning seed. The right shows the two fresh variance observations and their descriptive mean, not a confidence interval.

Construction	Settings	Mean V	Sample s	Mean Vt
Fixed	$\lambda = .1$	7.305e-4	.210	1.537e-4
Fixed, selected	$\lambda = .4$	1.683e-5	.217	3.640e-6
Polynomial	$L = .1, R = 1$	1.439e-3	.726	1.045e-3
Polynomial	$L = 1, R = 1$	1.015e-5	.781	7.944e-6
Polynomial	$L = 10, R = 1$	1.494e-3	.817	1.220e-3
Polynomial, selected	$L = 1, R = .25$	5.072e-6	.751	3.802e-6
Sign-aware, selected	$L = 1, R = .25$	5.072e-6	1.049	5.314e-6

All dynamic rows use $S = .8$. V is the trace of the finite estimator covariance. Times exclude loading/pilot and the measured coordinate-hashing/top-point retention spans. Vt is averaged per run, not formed from a pooled estimate. It is a short-run efficiency diagnostic, not an asymptotic convergence law.

8.7 What the largest-weight points show

The selected dynamic choices reduce confirmed variance by **3.318×** relative to tuned fixed $\lambda = .4$. Their individual variances are $[4.134, 6.010] \times 10^{-6}$, versus fixed $[1.633, 1.733] \times 10^{-5}$. The polynomial runtime cost offsets this gain: its mean Vt is 1.044 times fixed, while sign-aware is 1.460 times fixed. There is **no demonstrated sampling-speed advantage** in this small comparison. Including diagnostic instrumentation in the timer would misleadingly favor the slower method, because the same bookkeeping is a larger fraction of the fixed evaluator's cost. Separately measured setup and sampled RSS remain in the native evidence.

The large-cap deterioration is reproducible on the fresh seeds. To investigate it, the union of recorded largest-weight points was replayed under every cap, separately from production timing. There are 282 replayed points per dynamic construction and 66 same-arm full-vector parity checks each. No samples or new estimates enter this diagnostic. Symbolica supplies derivatives/evaluators; the native radius solver and matrix determinant supply the remaining analysis. Diagnostic strengths agree with the actual callback strengths to within 1.8×10^{-15}.

For example, at one identical recorded point in residual kernel 10 (source chart 15, shift 7, point 431), polynomial deformation gives:

L	$\lambda(x)$	$\lvert F(z)\rvert$	$\lvert\det J\rvert$	Weighted real output
.1	.0800	1.087	1.040	.1722
1	.7989	1.088	5.322	−.7322
10	6.728	1.201	820.8	−108.94

The magnitude $|F(z)|$ does not approach zero at this point as the cap increases; the map strongly amplifies the density. The full Jacobian is smaller than the frozen-strength diagnostic determinant at $L = 10$ (821 versus 1194). Thus this evidence does not support blaming only the rank-one $v(\nabla\lambda)^T$ term, and does not justify dropping it. Other recorded maxima in kernels 5, 7 and 9 remain over 99.2% cap-dominated at $L = 10$, with determinant amplification as the cap grows. These observations explain specific large weights; they are not a proof that one point determines the whole covariance.

At the chart center the displacement constraint can already dominate, so a center-only probe would miss this behavior. The four nonnegative shares of H and the actual weighted points distinguish these regimes. Reducing R from 1 to .25 at $L = 1$ improves the confirmed polynomial variance by about 2×; increasing it to 4 worsened the tuning result. Increasing S from .8 to .95 also worsened that result. Causal freedom therefore needs variance tuning. The existing runtime parameters provide a measurable improvement without a new production ansatz.

A possible later extension is a nonnegative smooth envelope term $\theta r^2 \|Dv\|_F^2$, with $\theta \geq 0$. It preserves the existing sufficient causal bound and would constrain $\lambda\|Dv\|_F$. It does not bound $v(\nabla\lambda)^T$ and has not been implemented or benchmarked. Any such extension must retain all symbolic derivatives, native checker consistency, and artifact identity; the present results make no claim for its performance.

8.7.1 Reproducible small source selections

The maintained kite TOML card accepts `generation.source_sectors = [2, 5]`. These are original geometry-chart IDs before symmetry and subtraction; this pair produces one symmetry-weighted numerical kernel. Native metadata saves the original total and local-to-original mapping. Inspection and numerical results label the selected contribution, including after portable/indexed reload. Omit the option for the full integral. This permits one- or two-sector generation experiments without confusing them with complete amplitudes.

Full native estimates, covariances, stream hashes, timings, generation sizes, pilot failures and settings are retained in `contour-cap-scalar-results.json`; the independent review is `reviews/contour-cap-ansatz-study.md`.

8.8 Physical double-box: increasing the saved cap

The physical D05 double-box at $\sqrt{s} = 1000$ GeV is reused without symbolic generation or optimization. Its historical programs have **Horner zero**: fixed uses a symbolic deformation Jacobian, polynomial dynamic uses the contour-only dual Jacobian, and both retain symbolic endpoint IBP. This is a runtime cap study of those existing programs, not a newly generated Horner-ten comparison. All quantities below refer to this one diagram with the earlier normalization, not the complete $gg \to hh$ amplitude.

The earlier 50-core campaign used $L = 10^{-6}, S = .8, R = 1$, hence $\lambda(x) \le 8 \times 10^{-7}$. A scout now scans $L = 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, .01, .1, 1, 10, 100$ at $S = .8, R = 1$. It retains numerical sectors `[0,3,6,11,26,28,29]`, with exact offsets explicitly excluded, and 1024 points × eight shifts per sector. These are partial-integral estimates on one paired tuning seed.

Scout prescription	V	**Interpretation**
Fixed $\lambda = 10^{-6}$	1049.39	Historical fixed control
Dynamic $L = 10^{-6}, S = .8$	2529.35	Historical dynamic control
Dynamic $L = 10^{-5}, S = .8$	1376.14	Best cap in the first scan
Dynamic $L \ge .01, S = .8$	337343	Saturation through cap 100
Dynamic $L = 10^{-5}, S = .4$	213.565	Selected after the R/S scan

Here V is the trace of the finite complex estimator covariance. A subsequent nine-arm scan varies $S = .4, .8, .95$ and $R = .01, .1, 1, 10$ in the predeclared combinations retained in the evidence. The selected $L = 10^{-5}, S = .4, R = 1$ minimizes both V and its product with measured evaluation time. The two repeated controls reproduce their original complete vectors and covariance exactly. Poor admitted results remain visible: $L = 1, S = .8, R = .01$ gives $V = 5.484 \times 10^8$ on this same partial scope.

8.8.1 Why a much larger cap is not automatically better

All 198 largest-weight cross-cap replays pass, including 63 original-vector checks. At a recorded sector-29 point, raising the cap to 100 yields only $\lambda(x) = 2.528574 \times 10^{-4}$. The causal, displacement, positive-factor and cap shares of H are approximately .57728, .42252, .00021 and 10^{-11}. Increasing the cap cannot remove these other sufficient bounds. At that point the full determinant magnitude is 281409, versus 786384 with the strength derivative artificially frozen for diagnosis. Thus the observed large weight does not justify omitting the dynamic derivative; it already reduces the determinant. Base-chart attribution and the complete residual's other subtraction-face callbacks are recorded separately.

The useful setting therefore increases L tenfold while reducing S. This permits larger local strengths than the original small cap while avoiding some of the amplification from the larger safety fraction. It is a measured choice for this diagram, not a universal deformation default.

8.9 Full double-box confirmation on fresh samples

The three prescriptions were frozen before seeds `202610104201` and `202610104202`. Each run covers all 30 six-dimensional residual kernels, 4096 points × eight complete Kuo33002 shifts per sector, Korobov3, and the native exact offset once. Same-seed point/weight hashes match across all prescriptions; the two seeds have different streams. Native complete Laurent vectors and their covariance are retained, including shared-shift sector cancellations. Each result is separate: no pooling across seeds or settings.

Prescription	V: seed 1 / 2	Sample s	Eval. µs avg/max
Fixed $\lambda = 10^{-6}$	1817.77 / 1511.09	7.510	6.769 / 7.695
Old dynamic $L = 10^{-6}, S = .8$	3517.00 / 2575.89	27.239	26.690 / 44.160
Selected $L = 10^{-5}, S = .4$	521.39 / 335.11	27.194	26.692 / 42.222

Both dynamic rows have $R = 1$. The table averages sampling times and sector means over the two seeds; its maximum is the largest sector mean in either run. Sampling excludes loading, pilot, coordinate hashing and largest-point bookkeeping, but includes the integration coordinator. Evaluator timings include precision rescues. These are one-core-per-run matched-work QMC measurements, not replacements for the earlier 50-core, 300-second campaign.

Relative to fixed strength, selected dynamic reduces variance by **3.486× and 4.509×**. Relative to the old dynamic prescription, the reductions are **6.745× and 7.687×**. Mean separately measured variances are 1664.43 fixed, 3046.45 old dynamic and 428.25 selected dynamic. These descriptive averages are not a combined integration estimate or a statistical confidence interval.

The corresponding fixed-to-selected Vt ratios are **.964 and 1.244**; there is no consistent timing advantage across both runs. The mean products are 12500.88 fixed, 82916.75 old dynamic and 11645.59 selected dynamic. Thus tuning removes much of the earlier dynamic variance penalty, while the contour evaluation cost remains material. Two short confirmations do not establish an asymptotic convergence law or an optimum.

Seed	Fixed finite mean	Old dynamic finite mean	Selected finite mean
4201	$26.17 + 36.63i$	$17.85 + 18.88i$	$31.50 + 42.33i$
4202	$52.47 + 28.40i$	$42.52 + 19.65i$	$73.43 + 46.12i$

These short-run estimates remain noisy: their joint standard errors are $\sqrt{V}$ from the first table. They do **not** claim per-mille accuracy, and there is still no independent physical D05 reference at this point. The purpose is matched-work variance and cost comparison, with the earlier longer estimates retained independently.

All six arms pass Pilot16, with zero optional production causal checks and precision rescue enabled. Restore plus integrity verification takes 7.04–7.07 s fixed and 10.00–10.17 s dynamic; binding takes .006 and .13 s. Pilots take 14.27–14.31 s fixed and 9.19–9.23 s dynamic. Maximum sampled process-group RSS is 257.38 MB fixed and 636.97 MB dynamic, including setup. At most two independent invocations run concurrently; the complete batch closes in 122.58 s with 1.238 GB aggregate peak RSS and no limit hit. No new generation, MC sweep, or sign-aware double-box generation is claimed.

Full native estimates, settings, timings, covariance and work-identity evidence are retained in `contour-cap-double-box-results.json`. The independent scientific and ecosystem review is `reviews/contour-cap-ansatz-study.md`. Existing parameters suffice for this measured improvement; a new Jacobian-distortion constraint remains a proposal.

9 No-contour threshold decomposition: first profiles

9.1 Scope and generation cost

These are **complete small-input integrations** using the current `threshold_decomposition` option, measured on 11 October 2026. They exercise verified GCAD cells, regular algebraic boundary maps, causal phases, symbolic endpoint subtraction and saved native evaluators. General singular-boundary resolution remains under development. No no-contour double-box timing or convergence result is available in this checkpoint.

The graph bubble has $s = 16$, $m_1^2 = m_2^2 = 3$, $\mu^2 = 1$, $D = 4 - 2\varepsilon$ and native OneLOop-compatible r_Γ normalization. Both arms compute through ε^1. The fixed-contour comparator uses the **same** graph, kinematics, normalization and orders, with $\lambda = .1$ admitted by a 16-point causal pilot; optional production contour checks are disabled. Both arms retain tracked numerical stability and precision rescue.

The two algebraic controls integrate $(P - i0)^{-1-\varepsilon}$ over the unit square, through ε^0. Degree seven uses $P = y^7 + y - x$, with two cells and two stochastic kernels. The sextic uses $P = y^6 - y + \frac{1+x}{16}$, with three cells and four stochastic kernels. Root boundaries retain exact defining polynomials and selectors; radical formulae are unnecessary. These are regular-branch controls, not tests of unresolved root collisions.

Case	Kernels	Native s	CLI s†	RSS MB†	Saved kB	Exact / JIT kB
Bubble / no contour	6	0.436	0.500	169.44	76.96	4.40 / 7.84
Bubble / fixed	1	0.026	0.054	29.25	21.33	1.14 / 2.20
Degree seven	2	0.405	0.449	83.31	69.07	2.10 / 3.01
Sextic	4	0.420	0.486	129.09	250.35	7.05 / 10.77

All arms use release mode, eight physical cores of an AMD EPYC 9754, symbolic Taylor subtraction, SymJIT O2, Horner ten, CPE iteration cap 1000 and common-pair distance 1000. Native seconds are the original generation ledger total. † CLI wall and sampled aggregate RSS come from a separate identical profiling repeat. It includes loading the input, generation and publication; native totals have narrower scope. Saved size is the manifest plus indexed data file. Exact/JIT sizes sum the inspected stochastic evaluator programs, excluding metadata and machine code. Memory and size units are decimal. Each generation is measured once with each observer; these are observations, not timing distributions.

Native stage ledger, milliseconds. These fields are retained as emitted; parallel worker spans and coordinator time do not form an exclusive partition. The current ledger does not isolate GCAD solving, independent verification, endpoint certification or root evaluation into separate timers.

Case	Geometry	Map	Subtract	Laurent coeff.	Compile
Bubble / no contour	9.46	4.15	1.09	26.41	171.14
Bubble / fixed	0.72	0.82	0.00	4.50	3.53
Degree seven	5.15	2.54	0.32	8.16	142.94
Sextic	4.68	8.39	1.12	27.78	156.77

9.2 Evaluator cost and precision rescue

Case	Diagnostic µs avg/max†	QMC mean µs	Max sector µs	Rescued %
Bubble / no contour	0.199 / 0.228	18.64–18.84	24.98	47.92–47.93
Bubble / fixed	0.317 / 0.344	0.43–0.45	0.45	0.02–0.02
Degree seven	499.865 / 908.674	1345.97–1348.37	1626.32	71.04–71.06
Sextic	483.217 / 1802.692	1119.06–1133.21	1725.08	64.11–64.12

The diagnostic uses 256 interior points and three repetitions per kernel. † Its average is the unweighted mean of kernel means; its maximum is the largest repetition mean, not the slowest individual point. QMC columns use both final-budget runs: range of mean sector costs, and the largest sector mean across either run. These operational costs include amortized preparation and precision rescue; they are not integration wall time. All sectors receive equal work here, so the pooled and unweighted means coincide.

The high-degree cases exercise numerical root evaluation and substantial arbitrary-precision replay. Their checked sampling costs are currently around one millisecond per sector point. The ledger does not identify how much is root solving versus precision replay. Certified branch geometry and tracked whole-vector evaluation do not amount to an interval certificate for every sample or the integrated result. No failed or cutoff-zero evaluations occur.

9.3 Integration time, memory and final-coefficient SD

Democratic QMC uses Kuo33002, Korobov3 and eight independent shifts. Each row below uses 4096 points **per shift per stochastic kernel** and eight workers. A/B are independent seed families, ending in 3701/3702 at this budget. Every run completes its allocation. Accepted work counts actual sector-point evaluations: six bubble threshold kernels cost six times the work of the one-kernel contour arm at the same per-kernel lattice size.

Case	Seed	Accepted work	Native s†	Load ms†	CLI s†	RSS MB†
Bubble / no contour	A	196,608	0.5143	6.13	0.5405	26.80
	B	196,608	0.5068	6.42	0.5332	27.00
Bubble / fixed	A	32,768	0.0052	2.93	0.0306	25.44
	B	32,768	0.0062	2.83	0.0321	26.38
Degree seven	A	65,536	12.9247	6.22	12.9510	27.66
	B	65,536	12.9370	6.42	12.9657	27.42
Sextic	A	131,072	25.6788	21.32	25.7221	30.59
	B	131,072	19.4868	20.92	19.5299	30.28

† All timing and RSS columns in this table come from identical profiling repeats, with bit-identical full estimates and covariance to the original scientific runs. Native integration time includes driver initialization, worker setup and checkpointing. Artifact loading is separate. The causal pilot occurs after loading and before that integration clock. CLI time spans process launch through observed exit, including loading, pilot, result saving and reporting. These clocks must not be added or described as pure sampling.

The faster observer targets 5 ms samples of parent plus descendant RSS, usually observing 5–7 ms intervals. The sextic A repeat also records a 4.24-second observation gap; its longer elapsed time is retained without adjustment. Shared pages may be counted in more than one process; sequential reads and missed short peaks prevent interpretation as unique memory or an exact simultaneous high-water mark. The earlier coarse observer missed brief compiler peaks; its measurements remain in the JSON, not in these tables.

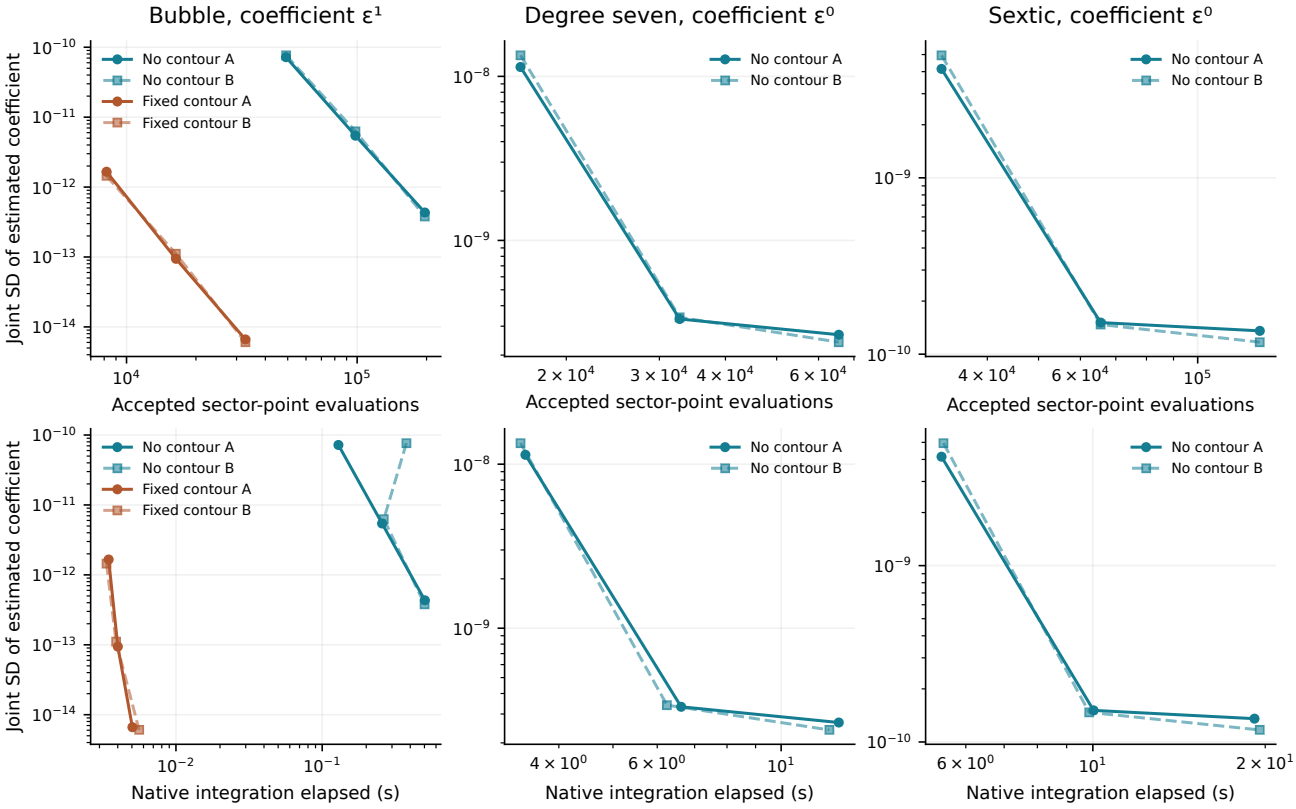
Last requested coefficient and its uncertainty. SD means the native standard deviation of the **estimated mean** (standard error), derived from complete independent QMC shifts. It is not the standard deviation of raw integrand values. Joint SD is $\sqrt{\operatorname{tr}\Sigma_k}$ for the real/imaginary covariance block of coefficient I_k, not a confidence radius.

Case	Seed	eps	Real	Imaginary	SD Re / Im	Joint SD
Bubble / no contour	A	1	−0.24374691221	0.96400656329	4.291e-13 / 5.637e-14	4.328e-13
	B	1	−0.24374691221	0.96400656329	3.762e-13 / 5.022e-14	3.795e-13
Bubble / fixed	A	1	−0.24374691221	0.96400656329	3.204e-15 / 5.800e-15	6.626e-15
	B	1	−0.24374691221	0.96400656329	2.933e-15 / 5.297e-15	6.055e-15
Degree seven	A	0	−0.02689338676	2.50241789116	2.591e-10 / 6.060e-11	2.661e-10
	B	0	−0.02689338657	2.50241789119	2.342e-10 / 5.146e-11	2.398e-10
Sextic	A	0	−2.81732392199	3.85211604612	8.176e-11 / 1.082e-10	1.356e-10
	B	0	−2.81732392203	3.85211604617	7.060e-11 / 9.330e-11	1.170e-10

Displayed means are rounded for readability; the JSON preserves every native digit, all Laurent coefficients and full covariance matrices. All 24 original runs report full-integral scope, available uncertainty and the requested accuracy stop (10^{-12} absolute, 10^{-8} relative on the last coefficient). This numerical stop is not an independent proof of accuracy.

Independent controls. The bubble pole is one and its finite coefficient is $2 - 3\frac{\log(3)}{2} + i\frac{\pi}{2}$. Both arms agree with this reference; their ε^1 comparison has no independent integrated analytic oracle. The polynomial poles cancel exactly in the retained vectors. The degree-seven finite imaginary part is $\pi r = 2.5024178911883874$, where $r^7 + r = 1$; the sextic reference is 3.8521160461762443 from the endpoint root-gap identity. No independent finite-real reference is supplied for these two polynomial controls. Reference residuals and statistical uncertainties remain distinct quantities.

9.4 Convergence versus accepted work and native elapsed time



Each line is one seed family; markers show independent lattice sizes 1024, 2048 and 4096. Lighter dashed lines are the second family. Curves use the **original** native integration-driver times, not the profiling repeats. Estimates and covariance are never pooled across seeds or lattice sizes. The work axis counts all accepted sector points. Shared seeds between the bubble arms do not supply cross-arm covariance: a calibrated significance for their difference is not inferred.

The fixed-contour bubble is cheaper and has smaller reported uncertainty at these settings. Its largest-budget SD is already near floating-point roundoff; that value must not be treated as a validated error bound or extrapolated scaling law. The no-contour bubble reaches joint SD about 4×10^{-13}, with a larger sector count and substantial precision rescue.

The higher-degree examples have low reported uncertainty but expensive checked sampling. Their largest-budget joint SD is about 2.4–2.7×10^{-10} (degree seven) and 1.2–1.4×10^{-10} (sextic). The improvement slows between the final two budgets in these observations. Three small budgets and two seeds do not establish an asymptotic rate or calibrated uncertainty coverage. No fixed/dynamic contour comparison is measured for these polynomial controls.

9.5 Reproduction and outstanding work

The frozen scientific source is `20bb812e31bc09a8a28a122d914a1de75e92cae7` on `no_deformation`; the report adds no sampling implementation changes. The optimized executable SHA-256 begins `7f41b135fdde78d8`. Complete settings, source and artifact identities, per-sector costs, precision diagnostics, native vectors and covariance are recorded in `docs/no-contour-results.json`. Reproducible cards and commands are in `examples/no_deformation/benchmarks/`; the independent measurement review is `docs/reviews/no-deformation-benchmark-audit.md`.

This checkpoint measures ordinary generation/integration. It does not repeat the separate normal/serial correctness matrix. Native GCAD preparation and compiler workers are caller-owned. Generic singular endpoints, broader parameter families and the physical ggHH comparison remain delivery work. The following double-box study records initial parametric and exact-point trials, followed by a successful extended rational-point solve and independent verification. The exact-point geometry is now complete; a fully regularized physical integration artifact remains unfinished.

10 Massive sunrise: complete no-contour correctness control

This new control extends the no-contour implementation beyond the massless sunrise in the opening benchmark table. It uses three masses equal to one, $s = 16$, $D = 4 - 2\varepsilon$, and the complete native density $-\Gamma(2\varepsilon - 1)U^{-3+3\varepsilon}(F - i0)^{1-2\varepsilon}$. The genuine projective source retains its original Gamma prefactor and causal phase. All seven verified cells contribute through 62 closed endpoint charts.

Laurent coefficient	Real part	Imaginary part
ε^{-2}	$1.4999998621 \pm 1.87 \times 10^{-7}$	0
ε^{-1}	$-1.2316465861 \pm 4.39 \times 10^{-7}$	0
ε^{0}	$10.0161411847 \pm 1.72 \times 10^{-6}$	$-3.3089881510 \pm 6.71 \times 10^{-8}$

Fixed contour, dynamic contour and independent pySecDec give compatible complete Laurent vectors. All 18 component comparisons pass without an added absolute-error floor; the largest difference is 2.162 combined standard errors. The pySecDec finite reference is $10.0161430322940 - 3.3089881368544i$. The no-contour relative standard error of the complex magnitude is 1.55×10^{-7}, retaining the full real/imaginary covariance.

Preparation and execution controls. Fresh verification of the saved proof, supplied-map admission, symbolic continuation and local publication took 12.38 seconds with a 60 MiB observed process peak. This excludes the original GCAD solve. The saved universal archive is 1.60 MB. Cold restoration and integration use that archive alone, without loading the original graph or proof.

The single-worker unoptimized Eager integration completed 16 independent native HKKN-alpha3 shifts, each with 256 points per chart: 253,952 calls in 275.88 sampling seconds, with a 42 MiB observed process peak. Korobov3 and seed 202610124919 were fixed in advance. It recorded 133,764 precision rescues, up to 256 bits, and no failures, unstable points or cutoff zeros. All 16 complete replica vectors and their 36-entry covariance were independently checked. Earlier incomplete diagnostic allocations were not pooled.

An optimized Eager build repeated the same archive and points in **97.83848 sampling seconds** and **101.49132 outer seconds**, with a **24 MiB** observed peak. All non-timing payloads matched exactly. A genuinely issued **SymJIT O2** archive then sampled the same points in **97.32653 seconds** (**100.95468 outer**, **30.18 MiB** observed peak): no material timing improvement in this single comparison. All means and replica vectors agree; the largest covariance difference is 1.22×10^{-24}. The repeated points are correlated and are never pooled or counted as additional independent data.

SymJIT preparation took 3.94 seconds with a 33 MiB process peak, including saved-proof verification and publication of a new 2.05 MB archive, excluding the original solve. All 62 restored sectors retain their native SymJIT backend. The same 133,764 points required rescue (52.67%); the ten slowest sectors used 62.94 seconds, seven rescuing every point. These diagnostics do not establish the cause of rescue.

These clocks are not comparable to the optimized 25-core, approximately ten-minute arms in the overview, and no square-root time extrapolation is applied. Ordinary and serial CLI generation now produce all 62 charts, and a separate cold process restores one selected record. Their clocks are outside this control. The opening overview now adds a separate fresh 25-core no-contour campaign: complete CLI generation in 6.70 seconds, followed by independent QMC and ordinary discrete-MC confirmations near 600 seconds. It does not reuse the one-worker estimates above. Its compact record is `docs/no-contour-massive-sunrise-overview.json`; matching fresh contour timings remain pending.

The general algebraic resolver, mixed-mass kite and complete double-box no-contour results remain unfinished.

Compact results and exact settings are retained in `docs/no-contour-massive-sunrise-control.json`. The native no-contour CLI card is `examples/no_deformation/massive_sunrise.toml`; the contour reference card is `examples/no_deformation/benchmarks/controls/massive_sunrise.toml`.

10.1 Physical double box: initial GCAD feasibility trials

A follow-up tests the same physical D05 graph at $\sqrt{s} = 1000$ GeV before attempting full no-contour generation. Native HEPKit/FastSecDec exports its exact Symanzik geometry in about 0.2 seconds. It retains the graph and numerator data, but these probes perform no numerator contraction, endpoint resolution or integration. The fully symbolic export has 15 terms in U and 80 in F.

The first solve keeps only s, t parametric, with exact masses $m_H = 125$ and $m_t = \frac{345}{2}$. Its open physical chamber includes $999000 < s < 1001000$, $-97100 < t < -97050$, $s > 4m_H^2$, $s > 4m_t^2$ and $t(2m_H^2 - s - t) > m_H^4$. Free mass parameters were not tested.

The fallback specializes **all geometry coefficients rationally**:

$$s = 1000000, \quad t = -97077, \quad u = -871673, \quad m_H^2 = 15625, \quad m_t^2 = \frac{119025}{4}.$$

This differs slightly from the original $t = -484375 + 100000\sqrt{15}$ at $\cos(\theta) = \frac{4}{5}$. The new $\cos(\theta)$ is approximately 0.79999930881; on-shell conditions, momentum conservation and Gram products pass independent native exact checks. Any later amplitude comparison must use this same point and regenerate its external polarization data.

Geometry / order	Solve wall s	OS RSS MiB	Outcome
Parametric s, t, baseline	60.037	179.63	Projection timed out
Exact rational point, baseline	180.043	198.84	Projection timed out
Exact rational point, alternate	180.037	169.48	Projection timed out

These three initial scouts are single-worker native solver probes, not complete generation times. RSS is the child process's OS-reported maximum from `wait4`; separate periodic RSS observations are retained in the JSON. Build time and U/F export are excluded. All three stops are imposed by the external wall deadline. The process address-space bound is 10 GiB; observed memory is far below it. Configured native time/memory settings do not cap eager projection in this direct library path. An external supervisor enforces the stated limits; raising the memory allowance alone would not address these observed stops.

Every trial uses symGCAD's optimized engine, positivity pruning and its checked homogeneous reduction of the positive $x_0, \ldots, x_6$ domain with anchor x_0. The baseline requests coordinate lifting order; the alternative uses $[x_0, x_5, x_4, x_3, x_2, x_1, x_6]$. Parameters precede integration variables. The alternative is selected by a native cheap ordering heuristic, with no unreported trial projections. The final completed projection records contain 15 polynomials / 406 terms for the parametric case, 22 / 1393 for the fixed baseline and 25 / 1643 for its alternative. These are partial diagnostic records, not verified cells.

None of these initial trials returned a SolveResult. They do not prove mathematical infeasibility. The next page reports the successful independently verified rational-point geometry obtained with a different native prescription and a larger allowance. That result is separate from this initial table; endpoint resolution and integration are still required afterward.

The runner links release-built public symGCAD `a1132d4f` and Symbolica/Numerica `ae0c82b7`; its small standalone driver uses opt-level zero and thin LTO. It is a separately labelled feasibility build, not the release CLI executable behind the small-input tables. Exact settings, dependency identities and measurements are in `docs/no-contour-gghh-feasibility.json`. The native source/algorithm can be replayed from `examples/no_deformation/gghh_feasibility/`. No later upstream production solver changes were found when checking symGCAD head `ea9d1cf5`.

10.2 Physical double box: verified rational-point geometry

With a one-hour solve-plus-verification budget and a 100 GB allowance, the **complete fixed-point GCAD problem now succeeds**. The point and physical graph are those on the preceding page. Native symGCAD retains the entire positive orthant and both signs of F; no negative-only region or selected sector is substituted for the full geometry.

A projection-free scan of all six ratio variables admits one native conditional-quadratic elimination, in x_2. Its exact positive-coefficient certificates hold on the complete domain. The successful native prescription is positive homogeneous reduction at x_0, the monotone engine, and order $[x_0, x_1, x_3, x_4, x_5, x_6, x_2]$, with the optimized engine for the remaining problem. Solver and dependency revisions are unchanged. The original optimized order separately reached its one-hour limit without completing.

Completed geometry measurement	**Result**
Verified original generic-open cells	644,699
Native solver total	36.356 s
Native projection / lifting	0.446 / 23.186 s
Solve process, including saved raw proof	156.526 s
Fresh verification process, including reload	142.072 s
Complete geometry trial wall time	298.601 s
Maximum child OS peak RSS	10.688 GiB
Maximum sampled aggregate cohort RSS	11.875 GB
Saved raw native proof	5.284 GB
Full integrand generation / integration / SD	Pending

The helper saves the raw result before starting a verifier in a fresh process. Exact input, effective order and parameter-prefix association pass, as does native independent verification of all 644,699 original cells. The reduced base's 464,265 cells are a separate counter. Its solver total includes more than projection and lifting; those two stage times must not be summed and labelled as total generation.

Serialization is substantial: the solve helper's timer is 152.310 seconds including its raw JSON write, versus 36.356 seconds in native solver statistics. The process wall includes startup and teardown. Verification includes decoding the saved proof. Both child processes have one Rayon thread; they are pinned to CPUs 18–19. These measurements use the same separately identified Rust driver over release native libraries as the geometry probes, not the release CLI integration campaign.

Child RSS comes from `wait4`. Aggregate RSS is sampled across the original extended baseline's supervisor/child and the successful successor's owned processes; it excludes unrelated host work. The actual enforcement is 100 GB decimal address space per child and a 100 GB aggregate RSS watchdog. The native 65536 MiB setting is recorded separately. Neither external memory nor the pooled 3600-second deadline is reached by this successful trial.

Native source binding retains the actual 156-term physical numerator; 604,005 cells have an infinite boundary. The first cell now has an executable open map with two normalized algebraic sections and three compactified rays. All 42 coordinate/Jacobian comparisons at 128/256 bits pass independent root enclosures. Map preparation takes 0.058 s; this separate correctness run, including proof reload and verification, takes 258.725 s and peaks at 11.528 GB aggregate RSS. It does not time a compiled physical integrand. The next page records the subsequent full-density and infinity-chart checks. See `docs/reviews/no-deformation-normalized-maps.md`.

This completes the fixed-point generic-open geometry gate, not the physical integration. General cell maps, closed faces, causal zero sections, infinity boundaries and exceptional seams still require treatment. Native compatibility merges are not silently substituted for certified cells. Consequently there is no double-box no-contour integration runtime or SD yet.

The exact recipe is in `examples/no_deformation/gghh_feasibility/fixed-monotone-prepared.toml`; compact stage measurements, raw-proof digest and scope are in `docs/no-contour-gghh-verified.json`. Large raw proofs remain untracked. The rational-point Rust exporter now also supplies consistent physical momenta and polarization products for a later matched amplitude comparison.

10.3 Physical double box: density and infinity preparation

The same exact physical input now passes an open-density check on original cell zero, using all 156 source terms and four native infinity charts. The complete 644,699-cell proof is loaded and verified once. Native evaluator composition keeps the numerator in its original coordinates; the cell map feeds it without expanding the algebraic roots inside the numerator.

Preparatory correctness measurement	**Result**
Open cell map preparation	0.052 s
Four-chart endpoint transport	0.247 s
Complete 156-term open-density preparation	0.607 s
Native exact density IR	58,933 bytes
Term / summed-density comparisons	6,864 / 44; all pass
Independent coordinate/Jacobian comparisons	154; all pass
Proof decoding / source-bound verification	126.148 / 116.082 s
Process wall time, including teardown	254.308 s
Sampled aggregate peak RSS	11.522 GB
Full integrand generation / integration / SD	Pending

The density checks use real epsilon values 1/8 and 1/4 at 128/256 bits. A separate evaluator uses the original source factors, exponents and complex numerator, with an independent exponential representation of the lower-lip phase. Each chart's additional Jacobian appears once. The exact evaluator codec roundtrip retains its live algebraic-root owner; detached restoration is a separate gate. These are ordinary high-precision arithmetic comparisons, with independent directed-ball checks of the coordinates and Jacobian.

This correctness driver uses the private combined FastSecDec build and captured development dependencies. It is **not a release SymJIT sampling benchmark**. The 1.931 s numerical-check stage also constructs reference/composed evaluators and writes evidence; dividing it by the point count would not give sampling cost. The test uses one process under 600 s / 100 GB limits. Its shared proof owner establishes no cross-process residency bound.

For numerator behavior at the common infinity face, native finite-depth series completes all **128 distinct numerators × four charts** in **20.602 s**, at **38,588 KiB** child peak RSS. Twelve independent exact specialized coefficient controls pass. Joining the original powers and measure gives c^3 for all 624 term/chart combinations, with epsilon cancelling. This calculation avoids materializing the fully cleared numerator: an earlier expansion trial reached roughly 9 MB for one 12 KB source and was stopped at 265.8 s / 6.65 GB.

Endpoint regularization and full integration remain unfinished. The common infinity calculation does not resolve direction faces, lower $U = F = 0$ strata, finite causal seams or all other original cells. It supplies exact local coefficients, never a truncated bulk approximation. No missing contribution is treated as zero. Scope, source identities and compact measurements are in `docs/no-contour-gghh-verified.json`; source and reuse reviews accompany it.

Implementation checkpoint, `cf3cc91`. Complete replacement chart covers, original-factor obligations on every relative face, and compact symbolic numerator derivatives through moving algebraic roots are now implemented. Their maintained scientific checks, independent reviews and strict consuming workspace checks pass. Exact boundary zeros remain pending refinement; finite numerator coefficients are not a valuation or a complete density certificate. The physical preparation measurements above are unchanged. Full endpoint closure, common analytic continuation and integration remain required before a no-contour double-box SD or runtime comparison can be given.

11 Implementation and performance

11.1 Preserve algebraic structure and reuse native facilities

FastSecDec owns the sector/integration interfaces in Rust. Symbolica supplies factored atoms, differentiation, function maps, evaluator optimization, duals, and serialization; Numerica supplies arithmetic, matrices, RNG primitives and Havana. HEPKit and Linnet retain graph, model and kinematic ownership. HEPKit's Python bridge wraps native objects rather than duplicating algorithms.

The smooth density is deformed before subtraction. Native function definitions can share repeated Jacobian and envelope bodies while preserving their derivatives and explicit radius dependencies. The saved program restores the optimized evaluator; binding a strength or cap must not rerun Horner/CPE. SymJIT O2 remains the native backend default; portable exact evaluator IR is retained separately for eager and high-precision execution.

Horner defaults. New generation uses ten Horner iterations in the native, CLI and Python settings. The earlier double-box and scalar tables retain their explicit historical zero setting. In the pinned Symbolica implementation, zero takes the direct linearization path and skips the subsequent Horner and common-instruction optimization passes; it does not disable SymJIT O2. Contour-only dual composition performs additional native common-expression optimization separately. No historical program is relabelled as Horner ten.

An actual D05 source sector contains 236 smooth terms. The initial representation copied its large Jacobian into each term. Native structured construction plus a shared determinant body reduced that sector's mapped record from 683.8 MB to 365.6 MB and sampled peak RSS from 2.178 GB to 1.288 GB. Mapping wall time was 10.93 s versus 7.95 s on a shared host. In the native debug graph, compilation fell from 169.49 s to 58.16 s. Saved full-vector parity passes at two admitted points; both representations reject the same excessive strength. Saved program size increases by 6.6%. These are one-sector measurements, not full-generation timings or a convergence result.

11.2 Symbolic versus dual determinant

The symbolic route constructs a native determinant and optimizes it together with the density. It offers common-expression elimination across outputs but may make generation expensive when a large body is repeatedly substituted. The first optional dual route also applied numerical-dual subtraction to the whole density. It coupled the two choices, contrary to the clarified requirement. Its historical measurements include high-order endpoint jets and cannot establish the cost or limits of the corrected contour-only construction.

Independent contour-only construction. Endpoint reduction and subtraction remain symbolic in both variants. Native differentiation retains every bulk, boundary, determinant and local-strength derivative. At final compilation, native exact-subtree replacement supplies surviving first-image derivatives from a degree-one image-only Dualizer; higher endpoint derivatives are already symbolic. Coordinate seeds remain one on subtraction faces, and parameter seeds remain zero. Native pruning removes unused inputs before composition. Exact-only boundary contributions retain symbolic evaluation and cancellation.

Five focused compiler controls pass, including third-order endpoint poles with fixed and both dynamic prescriptions, saved programs, unused callbacks and private-input collisions. Public Symbolic IBP controls also pass with both implicit-radius prescriptions, complete complex Laurent vectors and restored programs. Conditional bodies requiring hoisted image inputs are explicitly unsupported by this opt-in route. The matched source-sector comparison below includes generation, RSS, program sizes and sampling cost. Full-integral generation and all four physical integration runs have completed. No mathematical reordering, alternative CAS or AD implementation is introduced. An optional initial native Series width avoids a known discarded attempt; the strict absolute Laurent-remainder coverage check and checked retries remain unchanged. It defaults to one. Twenty-three focused controls and independent native-reuse reviews pass. The measurements above distinguish generation savings from the cost of sampling the resulting evaluator.

11.3 Matched Jacobian measurements on one source sector

Contour / Jacobian	Time / peak RSS	Exact / JIT IR	Mean sample
Fixed / symbolic	559.9 s / 12.01 GB	108.3 / 156.2 kB	14.64 µs
Fixed / dual	566.9 s / 12.01 GB	136.8 / 202.8 kB	19.20 µs
Dynamic / symbolic	1511.6 s / 28.44 GB	294.0 / 524.6 kB	37.71 µs
Dynamic / dual	794.9 s / 16.73 GB	372.0 / 520.0 kB	29.99 µs

Times include generation, compilation, saving and pointwise checks in separate processes; sizes describe evaluator IR, not the complete saved owner. Settings are SymJIT O2, Horner zero and common-pair cap 1000. Each matched pair uses 8,192 identical points: fixed $\lambda = 10^{-6}$ and dynamic $S = .8$, $L = 10^{-6}$, $R = 1$. Both fixed arms rescue 27 points in

double-double; both dynamic arms rescue 28. Fresh-process restoration and complete-vector parity pass. This sector favors symbolic J for fixed mode and dual J for dynamic mode.

11.4 Residency, callback ownership and statistical work

Universal indexed artifacts let serial generation publish one completed sector and release its worker process. Selective loading retains optimized native programs. Ordinary discrete MC uses caller-owned parallel execution; the 50-core campaign must account for all resident evaluator contexts. Aggregate parent-and-child RSS measures actual residency, including JIT scratch.

Independent probes exposed a native JIT-clone callback issue: cloned evaluator tables were not used by machine code, which retained the original callback environment and serialized its mutable workspace. The narrow owner correction, Symbolica PR 62, is retained in public consumer revision `74225696`. The same consumer adds Symbolica PR 63: common-subexpression lookup remaps operands before hashing them, avoiding one optimizer pass per level of a duplicate dependency chain. Native branch ancestry and callback argument order remain unchanged. The preceding published workspace gate passed 935 tests (34 explicit ignores, including a child invoked separately), and its portable suite passed 83 tests. The source-selection follow-up passes 426 core tests (20 existing ignores), plus focused artifact, CLI process, portable selection and saved-result controls. Strict workspace Clippy, formatting and thin-binding checks pass. These are native/portable-host gates, not installed Python-wheel or browser/WASM execution evidence.

The physical QMC run also exposed avoidable coordinator work: matching common shifts used linear searches within already-sorted native replica records. Binary search preserves the selected rows and arithmetic order. Cancellation now drains dispatched results without repeating expensive observational reductions and checkpoint writes for every return; the final native checkpoint and report still include accepted returns. Deterministic cancellation and reordered, gapped-shift covariance/restoration controls cover these changes.

11.5 Persist the native primary evaluator

The old artifact saved exact evaluator IR but discarded the compiled primary. The new record also saves Symbolica's native JIT payload. Compatible loads restore its lowered application and callback descriptors; the owner still emits machine code. Exact IR remains available for precision rescue and portable execution. Existing mathematical identities and caller-owned scheduling remain unchanged. Immutable exact templates and artifact buffers are shared across workers; mutable stacks and callback environments remain independent.

On two preserved source-sector records, matched optimized cache-miss/cache-hit loads measured 0.1804/0.1437 s (fixed) and 0.3865/0.3025 s (polynomial). The optional cache adds 13.34% to those records. These are loader measurements on historical endpoint constructions, not new physical integration results. Changing dynamic validation factories also reuses the native primary instead of rebuilding it from exact IR. Optional byte-integrity checks remain optional; format, compatibility and native decoding checks are retained.

A second probe restores the same complete 70.41 MB fixed artifact in the old and new optimized implementations, then constructs 50 whole-owner worker clones. Post-clone RSS falls from 6.68 to 1.79 GB, and clone time from 3.32 to 0.78 s. Loading the original uncached artifact remains similar, 3.81 versus 3.71 s. This measures immutable ownership improvements without binding, pilot or sampling work; it is separate from the cache-hit comparison above.

Stream identities are coordinator-owned. Pilots, production replicas and retries keep separate identities; rejected, duplicate or stale work cannot enter the estimate. Optional causal checks are truly absent in unchecked production, while solver/nonfinite failures remain visible. Accepted common QMC shifts and complete MC batches update the full covariance, preserving sector cancellations even at a time-budget stop.

Reproduction and source: FastSecDec, contour_deformation. Native input: `examples/contour/gghh_double_box_1000/`. Mathematical audit: `docs/reviews/contour-foundation.md`. Physical campaign provenance and complete covariance are in `docs/contour-deformation-results.json`. The scalar protocol, complete vectors, covariance, work identities and measurements are in `docs/contour-scalar-results.json` and `docs/reviews/contour-scalar-benchmark-audit.md`. The maintained Rust driver and reproduction commands are in `examples/contour/scalar_benchmarks/`.